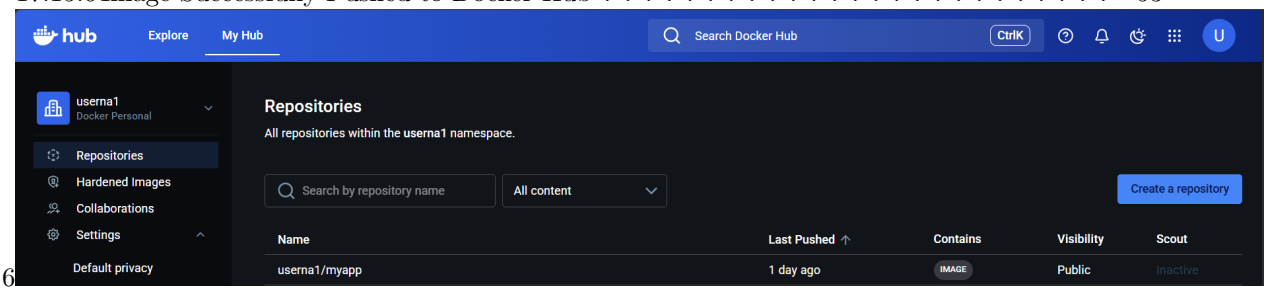
DevOps Lab Report

Tushar Mangal (500120513)

Contents



1 Containerization-and-DevOps

2 Containerization and DevOps

3 Name: Tushar Mangal

4 Sap Id: 500120513

4.1 About This Repository

This repository contains theory notes and lab experiments for the subject **Containerization and DevOps**. It focuses on understanding modern infrastructure concepts through both conceptual learning and hands-on practice.

Key areas covered: - Virtualization - Containerization - Docker - DevOps fundamentals

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4.2 Repository Structure

```
Containerization-DevOps/
|-- Theory/
|   |-- Class1/
```

```
|   |-- Class2/
|   |-- Class3/
|   |-- Class4/
|   |-- Class5/
`-- Lab/
    |-- Lab-1/
    |-- Lab2/
    |-- Lab3/
    |-- Lab4/
    |-- Lab5/
    |-- Lab6/
    |-- Lab7/
```

4.3 Theory Section

This section contains class-wise theory notes.

Link to theory index:

- Go to Theory README

Direct links:

- Go to Class 1 Notes
- Go to Class 2 Notes
- Go to Class 3 Notes
- Go to Class 4 Notes
- Go to Class 5 Notes

4.4 Lab Experiments

4.4.1 Lab 1 – Virtual Machines vs Containers

This lab demonstrates the comparison between Virtual Machines and Containers using: - VirtualBox - Vagrant - Ubuntu - Docker - Nginx

Lab documentation: Go to Lab 1 README

4.4.2 Lab 3

Deploying NGINX Using Different Base Images and Comparing Image Layers

Lab documentation: Go to Lab 3 Part1 README Lab documentation: Go to Lab 3 Part2 README

4.4.3 Lab 4

Lab documentation: Go to Lab 4 README

4.4.4 Lab 5

Lab documentation: Go to Lab 5 README

4.4.5 Lab 6

Lab documentation: Go to Lab 6 README

4.4.6 Lab 7

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4.4.7 Lab 9

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4.4.8 Lab 10

Lab documentation: Go to Lab 10 README

4.4.9 Lab 11

Lab documentation: Go to Lab 11 README

4.4.10 Lab 12

Lab documentation: Go to Lab 12 README

4.4.11 Assignment

Assignment - 1: Assignment 1

5 Containerization-and-DevOps

6 Lab Experiment 1 – Comparison of Virtual Machines and Containers

6.1 Objective

- Understand differences between Virtual Machines and Containers.
- Install and configure VM using VirtualBox and Vagrant.
- Install and configure Containers using Docker on WSL.
- Deploy Nginx web server in both environments.
- Observe system resource utilization.

6.2 Tools Used

- VirtualBox
- Vagrant
- Ubuntu
- WSL
- Docker
- Nginx
- VS Code

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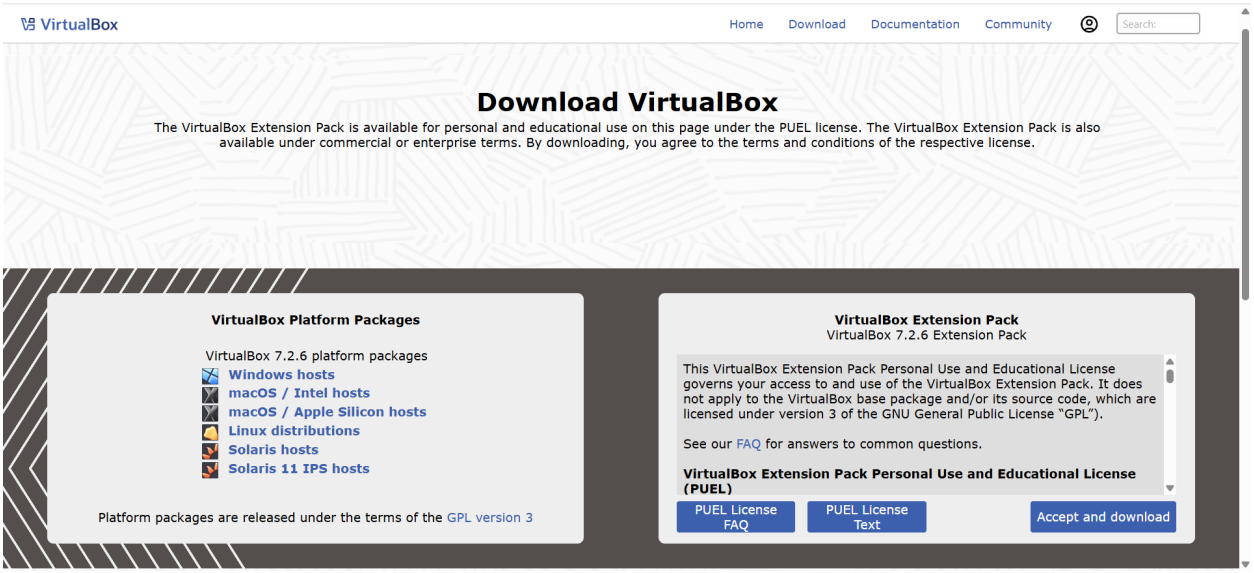
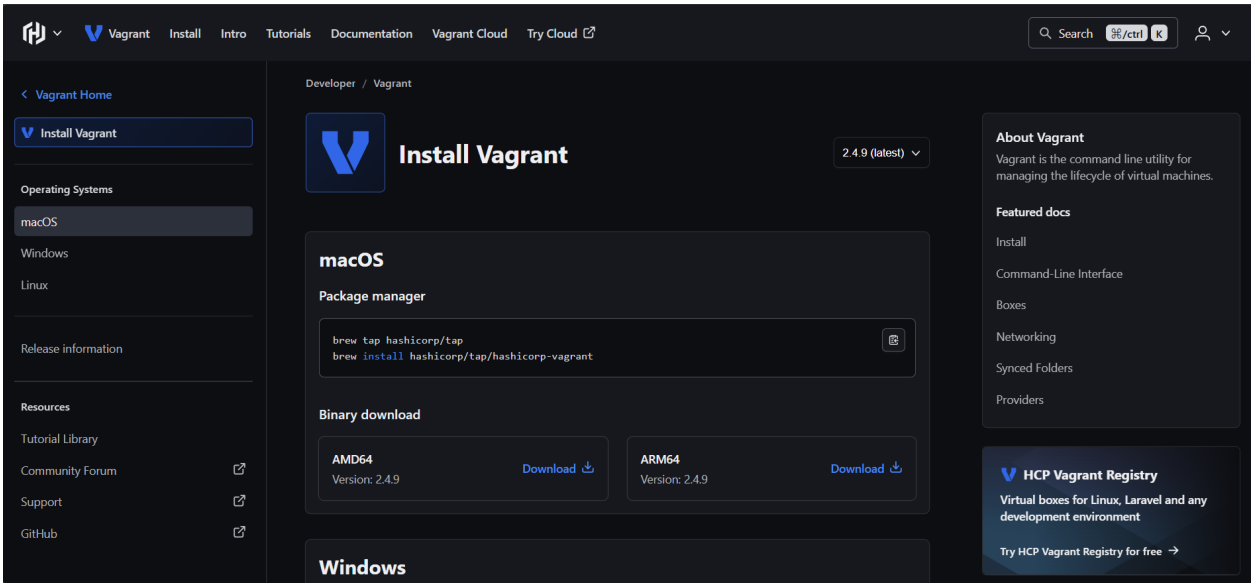
7 Part A – Virtual Machine Setup

7.0.1 Step 1 – Download VirtualBox



7.0.2 Step 2 – Install Vagrant



7.0.3 Step 3 – Initialize Vagrant

`vagrant init hashicorp/bionic64`

```
PS D:\semseter 6\lab> vagrant init hashicorp/bionic64
A `Vagrantfile` has been placed in this directory. You are now
ready to `vagrant up` your first virtual environment! Please read
the comments in the Vagrantfile as well as documentation on
`vagrantup.com` for more information on using Vagrant.
PS D:\semseter 6\lab>
```

7.0.4 Step 4 – Start VM

vagrant up

```
PS D:\semseter 6\lab> vagrant init hashicorp/bionic64
A `Vagrantfile` has been placed in this directory. You are now
ready to `vagrant up` your first virtual environment! Please read
the comments in the Vagrantfile as well as documentation on
`vagrantup.com` for more information on using Vagrant.
PS D:\semseter 6\lab>
```

7.0.5 Step 5 – VM Created

```
==> default: Clearing any previously set network interfaces...
==> default: Preparing network interfaces based on configuration...
    default: Adapter 1: nat
==> default: Forwarding ports...
    default: 22 (guest) => 2222 (host) (adapter 1)
==> default: Booting VM...
==> default: Waiting for machine to boot. This may take a few minutes...
    default: SSH address: 127.0.0.1:2222
    default: SSH username: vagrant
    default: SSH auth method: private key
    default: Warning: Connection reset. Retrying...
    default:
    default: Vagrant insecure key detected. Vagrant will automatically replace
    default: this with a newly generated keypair for better security.
    default:
    default: Inserting generated public key within guest...
    default: Removing insecure key from the guest if it's present...
    default: Key inserted! Disconnecting and reconnecting using new SSH key...
==> default: Machine booted and ready!
==> default: Checking for guest additions in VM...
    default: The guest additions on this VM do not match the installed version of
    default: VirtualBox! In most cases this is fine, but in rare cases it can
    default: prevent things such as shared folders from working properly. If you see
    default: shared folder errors, please make sure the guest additions within the
    default: virtual machine match the version of VirtualBox you have installed on
    default: your host and reload your VM.
    default:
    default: Guest Additions Version: 6.0.10
    default: VirtualBox Version: 7.1
==> default: Mounting shared folders...
```

7.0.6 Step 6 – VM Running

```
PS D:\semseter 6\lab> vagrant ssh
Welcome to Ubuntu 18.04.3 LTS (GNU/Linux 4.15.0-58-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

  System information as of Sat Jan 31 09:36:06 UTC 2026

  System load:  0.5               Processes:           92
  Usage of /:   2.5% of 61.80GB   Users logged in:     0
  Memory usage: 11%               IP address for eth0: 10.0.2.15
  Swap usage:   0%


0 packages can be updated.
0 updates are security updates.


vagrant@vagrant:~$
```

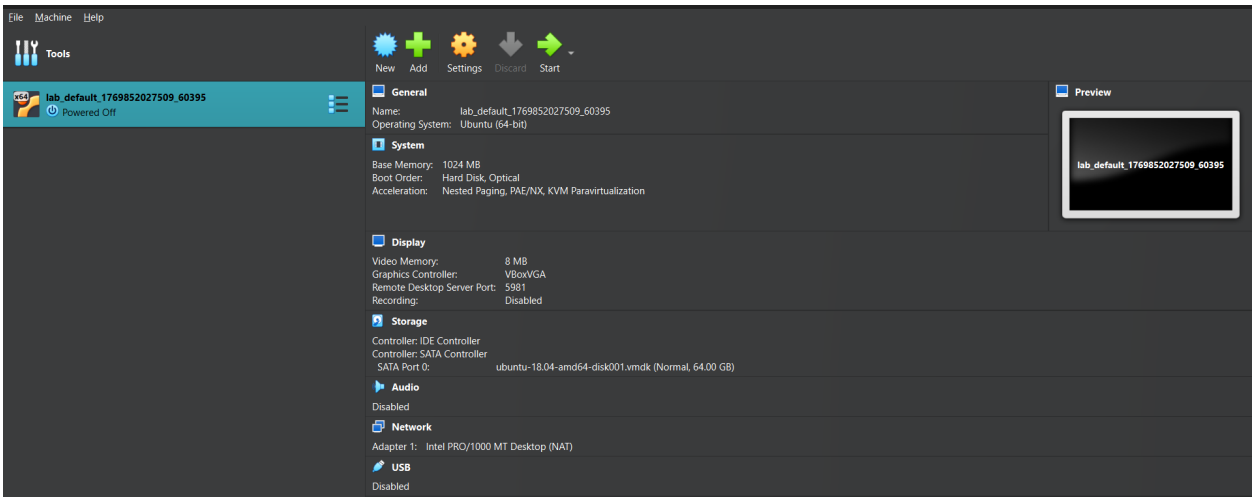
7.0.7 Step 7 – SSH into VM

vagrant ssh

```
vagrant@vagrant:~$ sudo systemctl start nginx
vagrant@vagrant:~$ curl localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
    body {
        width: 35em;
        margin: 0 auto;
        font-family: Tahoma, Verdana, Arial, sans-serif;
    }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
vagrant@vagrant:~$
```

7.0.8 Step 8 – Start Nginx in VM

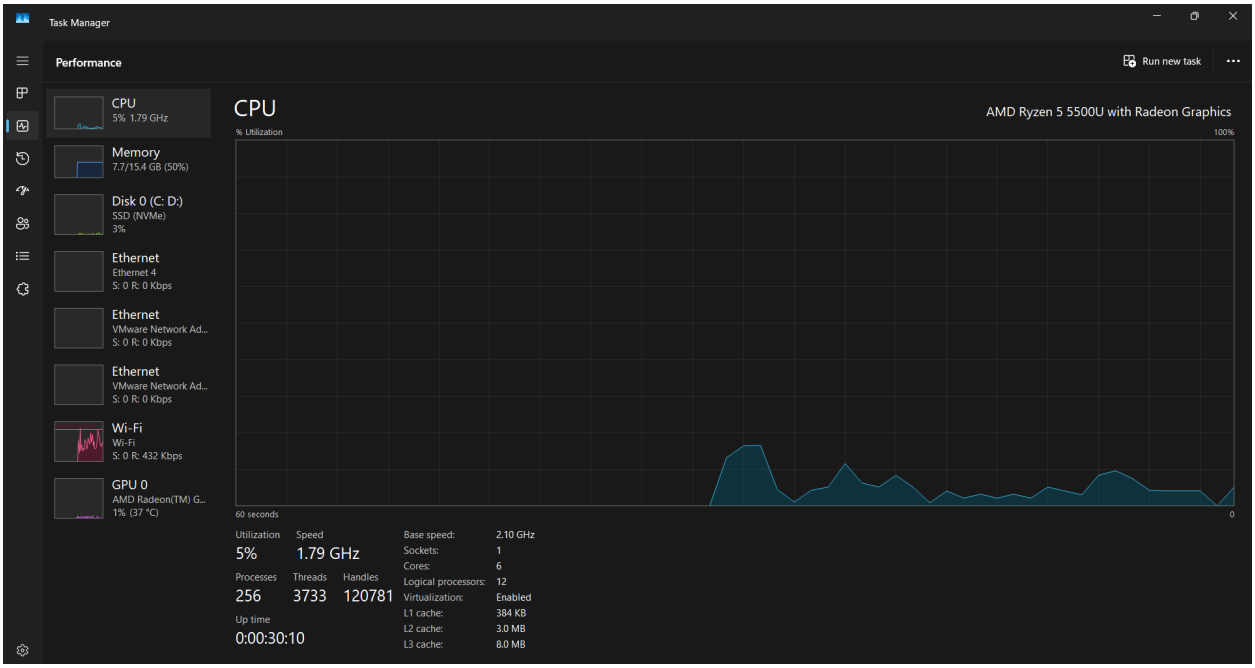
`sudo systemctl start nginx`



7.0.9 Step 9 – Verify Nginx

`curl localhost`

7.0.10 Step 10 – Virtualization Enabled



---

8 Part B – Docker Container Setup

8.0.1 Step 1 – Open WSL

```
PS D:\semseter 6\lab> wsl
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/d/semseter 6/lab$ docker start ubuntu
Error response from daemon: No such container: ubuntu
Error: failed to start containers: ubuntu
```

8.0.2 Step 2 – Run Docker Nginx

`docker run -d -p 8080:80 --name nginx-container nginx`

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/d/semseter 6/lab$ docker run -d -p 8080:80 --name nginx-container nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
119d43eec815: Pull complete
700146c8ad64: Pull complete
d989100b8a84: Pull complete
500799c30424: Pull complete
10b68cfefee1: Pull complete
57f0dd1befe2: Pull complete
eaf8753feae0: Pull complete
Digest: sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf4aec
Status: Downloaded newer image for nginx:latest
f67ef10ddcedbe2a775982a07aad6ee8568283bfc3df40031794a38676875442
```

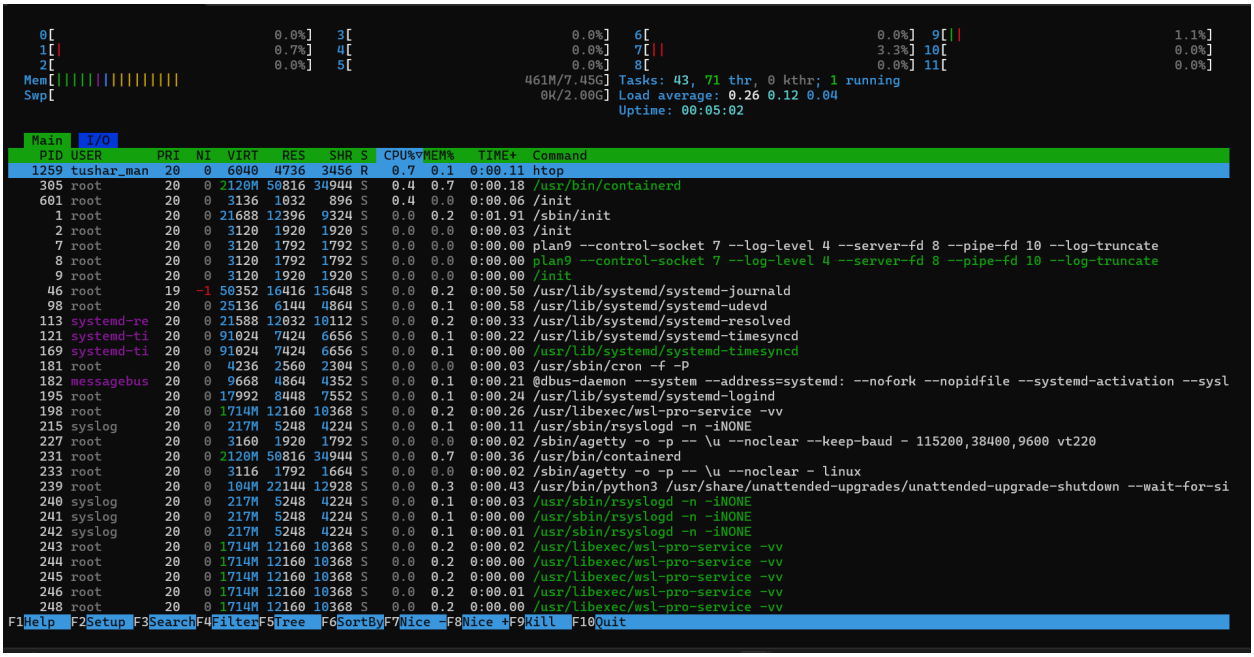
8.0.3 Step 3 – Verify Container Output

`curl localhost:8080`

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/d/semseter 6/lab$ curl localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

8.0.4 Step 4 – Monitor using htop

`htop`



8.0.5 Step 5 – Check Memory

`free -h`

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/d/semseter 6/lab$ free -h
               total        used        free      shared  buff/cache   available
Mem:           7.5Gi       626Mi       6.0Gi       3.6Mi       1.0Gi       6.8Gi
Swap:          2.0Gi          0B       2.0Gi
```

8.0.6 Step 6 – System Analyze

`systemd-analyze`

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/d/semseter 6/lab$ systemd-analyze
Startup finished in 5.856s (userspace)
graphical.target reached after 5.796s in userspace.
```

8.0.7 Step 7 – Docker Stats

`docker stats`

```
CONTAINER ID   NAME              CPU %     MEM USAGE / LIMIT    MEM %     NET I/O          BLOCK I/O     PIDS
f67ef10ddced   nginx-container   0.00%     10.43MiB / 7.45GiB   0.14%     1.72kB / 1.4kB   0B / 12.3kB   13
```

8.1 Result

Successfully compared VM and Container performance.

9 Containerization-and-DevOps

10 Experiment 4: Docker Essentials

10.1 Dockerfile .dockerignore tagging publishing

10.2 Part 1: Containerizing Applications with Dockerfile

10.2.1 Step 1: Create a Simple Application

Python Flask App:

```
mkdir my-flask-app
cd my-flask-app
```

```
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4> mkdir my-flask-app


    Directory: C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        21-02-2026     14:15                my-flask-app


PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4> cd my-flask-app
```

app.py:

```
from flask import Flask
app = Flask(__name__)

@app.route('/')
def hello():
    return "Hello from Docker!"

@app.route('/health')
def health():
    return "OK"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

app.py

```python
from flask import Flask
app = Flask(__name__)

@app.route('/')
def hello():
    return "Hello from Docker!"

@app.route('/health')
def health():
    return "OK"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)

```

requirements.txt:

```
Flask==2.3.3
```

requirements.txt

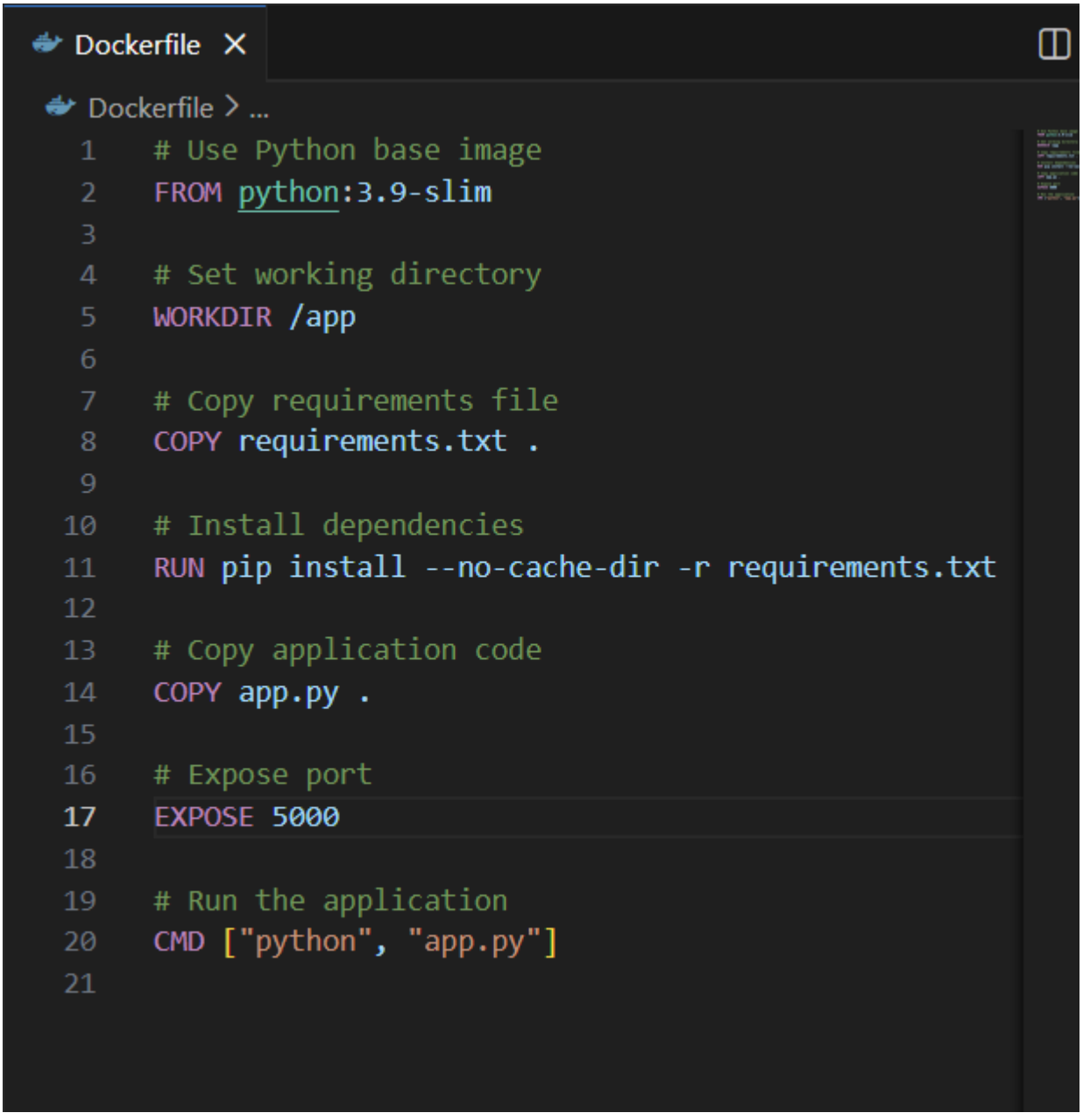
```
Flask==2.3.3

```

10.2.2 Step 2: Create Dockerfile

Dockerfile:

```
# Use Python base image
FROM python:3.9-slim

# Set working directory
WORKDIR /app

# Copy requirements file
COPY requirements.txt .

# Install dependencies
RUN pip install --no-cache-dir -r requirements.txt
```

```
# Copy application code
COPY app.py .

# Expose port
EXPOSE 5000

# Run the application
CMD ["python", "app.py"]
```

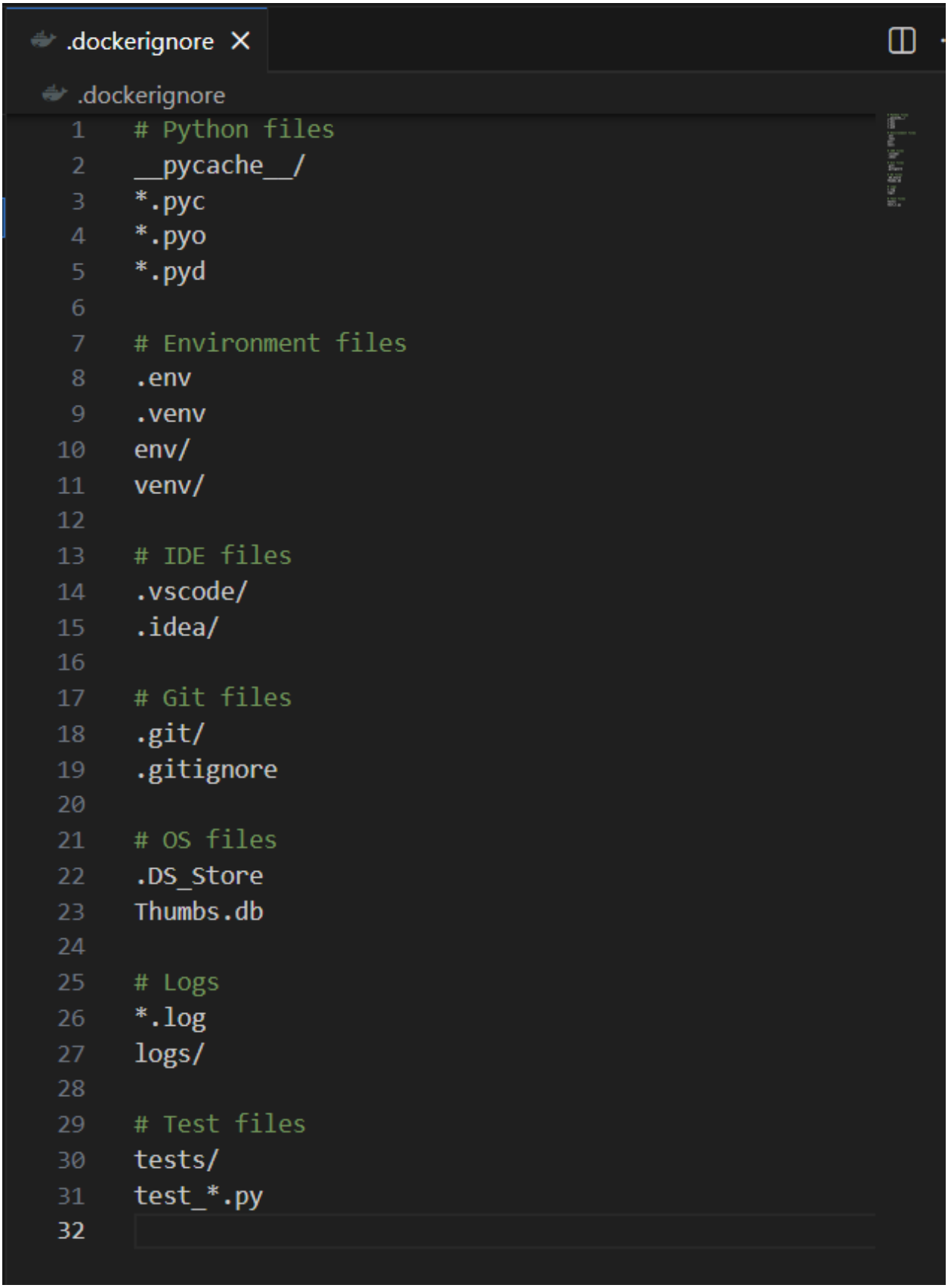


10.3 Part 2: Using .dockerignore

10.3.1 Step 1: Create .dockerignore File

.dockerignore:

```
# Python files
__pycache__/
*.pyc
*.pyo
*.pyd

# Environment files
.env
.venv
env/
venv/

# IDE files
.vscode/
.idea/

# Git files
.git/
.gitignore

# OS files
.DS_Store
Thumbs.db

# Logs
*.log
logs/

# Test files
tests/
test_*.py
```


.dockerignore
.dockerignore
Python files
__pycache__/
*.pyc
*.pyo
*.pyd
Environment files
.env
.venv
env/
venv/
IDE files
.vscode/
.idea/
Git files
.git/
.gitignore
OS files
.DS_Store
Thumbs.db
Logs
*.log
logs/
Test files
tests/
test_*.py

10.3.2 Step 2: Why .dockerignore is Important

- Prevents unnecessary files from being copied
- Reduces image size
- Improves build speed
- Increases security

10.4 Part 3: Building Docker Images

10.4.1 Step 1: Basic Build Command

```
# Build image from Dockerfile
docker build -t my-flask-app .

# Check built images
docker images
```

```
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/xwxm1kga08khgrgqtq5ctf2z3
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker build -t my-flask-app .
[+] Building 1.3s (10/10) FINISHED                      docker:desktop-linux
 => [internal] load build definition from Dockerfile                    0.0s
 => => transferring dockerfile: 381B                                    0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim      0.6s
 => [internal] load .dockerignore                                       0.0s
 => => transferring context: 304B                                       0.0s
 => [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16b   0.1s
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16b   0.1s
 => [internal] load build context                                       0.0s
 => => transferring context: 63B                                        0.0s
 => CACHED [2/5] WORKDIR /app                                           0.0s
 => CACHED [3/5] COPY requirements.txt .                                0.0s
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt     0.0s
 => CACHED [5/5] COPY app.py .                                          0.0s
 => exporting to image                                                  0.3s
 => => exporting layers                                                 0.0s
 => => exporting manifest sha256:c8d2616072f3cdd22c334d17834f3d624565   0.0s
 => => exporting config sha256:47bcef556a96a512e2e965ef7e584562a32fe1   0.0s
 => => exporting attestation manifest sha256:642430dc8c2f91124e563d74   0.1s
 => => exporting manifest list sha256:8300d23a75f1dc49dcb068b4992fff3   0.0s
 => => naming to docker.io/library/my-flask-app:latest                  0.0s
 => => unpacking to docker.io/library/my-flask-app:latest               0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/an34lbmcr6kgn7i3x5fb2g3ls
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker images
                                                    i Info →   U  In Use
IMAGE                  ID             DISK USAGE   CONTENT SIZE   EXTRA
cloud-robot:latest     1ae3bf81d146        857MB         191MB    U
my-flask-app:latest    8300d23a75f1        200MB        48.8MB
nginx-alpine:latest    3a47dbf1b178         16MB        4.74MB    U
nginx-ubuntu:latest    cb305251001b        199MB        52.7MB    U
nginx:latest           341bf0f3ce6c        240MB        65.8MB    U
```

10.4.2 Step 2: Tagging Images

```
# Tag with version number
docker build -t my-flask-app:1.0 .

# Tag with multiple tags
docker build -t my-flask-app:latest -t my-flask-app:1.0 .

# Tag with custom registry
docker build -t username/my-flask-app:1.0 .

# Tag existing image
docker tag my-flask-app:latest my-flask-app:v1.0
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker build -t my-flask-app:1.0 .
[+] Building 1.3s (10/10) FINISHED                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                  0.0s
 => => transferring dockerfile: 381B                                  0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim    0.6s
 => [internal] load .dockerignore                                     0.0s
 => => transferring context: 304B                                     0.0s
 => [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16b  0.1s
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16b  0.1s
 => [internal] load build context                                     0.0s
 => => transferring context: 63B                                      0.0s
 => CACHED [2/5] WORKDIR /app                                         0.0s
 => CACHED [3/5] COPY requirements.txt .                              0.0s
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt   0.0s
 => CACHED [5/5] COPY app.py .                                        0.0s
 => exporting to image                                                0.3s
 => => exporting layers                                               0.0s
 => => exporting manifest sha256:c8d2616072f3cdd22c334d17834f3d624565  0.0s
 => => exporting config sha256:47bcef556a96a512e2e965ef7e584562a32fe1  0.0s
 => => exporting attestation manifest sha256:bb49a0f6eea10a0a57978775  0.1s
 => => exporting manifest list sha256:c2771f6d4a602b095f232cc877ac064  0.0s
 => => naming to docker.io/library/my-flask-app:1.0                   0.0s
 => => unpacking to docker.io/library/my-flask-app:1.0                0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/sij6uo3tonwqjiw6kss2mmgm7
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker build -t my-flask-app:latest -t my-flask-app:1.0 .
[+] Building 1.2s (10/10) FINISHED                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                  0.0s
 => => transferring dockerfile: 381B                                  0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim    0.5s
 => [internal] load .dockerignore                                     0.0s
 => => transferring context: 304B                                     0.0s
 => [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16b  0.1s
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16b  0.1s
 => [internal] load build context                                     0.0s
 => => transferring context: 63B                                      0.0s
 => CACHED [2/5] WORKDIR /app                                         0.0s
 => CACHED [3/5] COPY requirements.txt .                              0.0s
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt   0.0s
 => CACHED [5/5] COPY app.py .                                        0.0s
 => exporting to image                                                0.3s
```

```
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/f6tjkqa8hnalv38pjhf8q74am
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker build -t username/my-flask-app:1.0 .
[+] Building 1.3s (10/10) FINISHED                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                  0.0s
 => => transferring dockerfile: 381B                                  0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim    0.6s
 => [internal] load .dockerignore                                     0.0s
 => => transferring context: 304B                                     0.0s
 => [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16b  0.1s
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16b  0.1s
 => [internal] load build context                                     0.0s
 => => transferring context: 63B                                      0.0s
 => CACHED [2/5] WORKDIR /app                                         0.0s
 => CACHED [3/5] COPY requirements.txt .                              0.0s
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt   0.0s
 => CACHED [5/5] COPY app.py .                                        0.0s
 => exporting to image                                                0.3s
 => => exporting layers                                               0.0s
 => => exporting manifest sha256:c8d2616072f3cdd22c334d17834f3d624565  0.0s
 => => exporting config sha256:47bcef556a96a512e2e965ef7e584562a32fe1  0.0s
 => => exporting attestation manifest sha256:561e974dcddbfc0ccf88fbe8  0.1s
 => => exporting manifest list sha256:678a8667bb828ccf8b6dca1c24b7d1a  0.0s
 => => naming to docker.io/username/my-flask-app:1.0                  0.0s
 => => unpacking to docker.io/username/my-flask-app:1.0               0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/y8xi8aica9wdly5pim1w8o0mu
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker tag my-flask-app:latest my-flask-app:v1.0
```

10.4.3 Step 3: View Image Details

```
# List all images
docker images

# Show image history
docker history my-flask-app

# Inspect image details
docker inspect my-flask-app
```

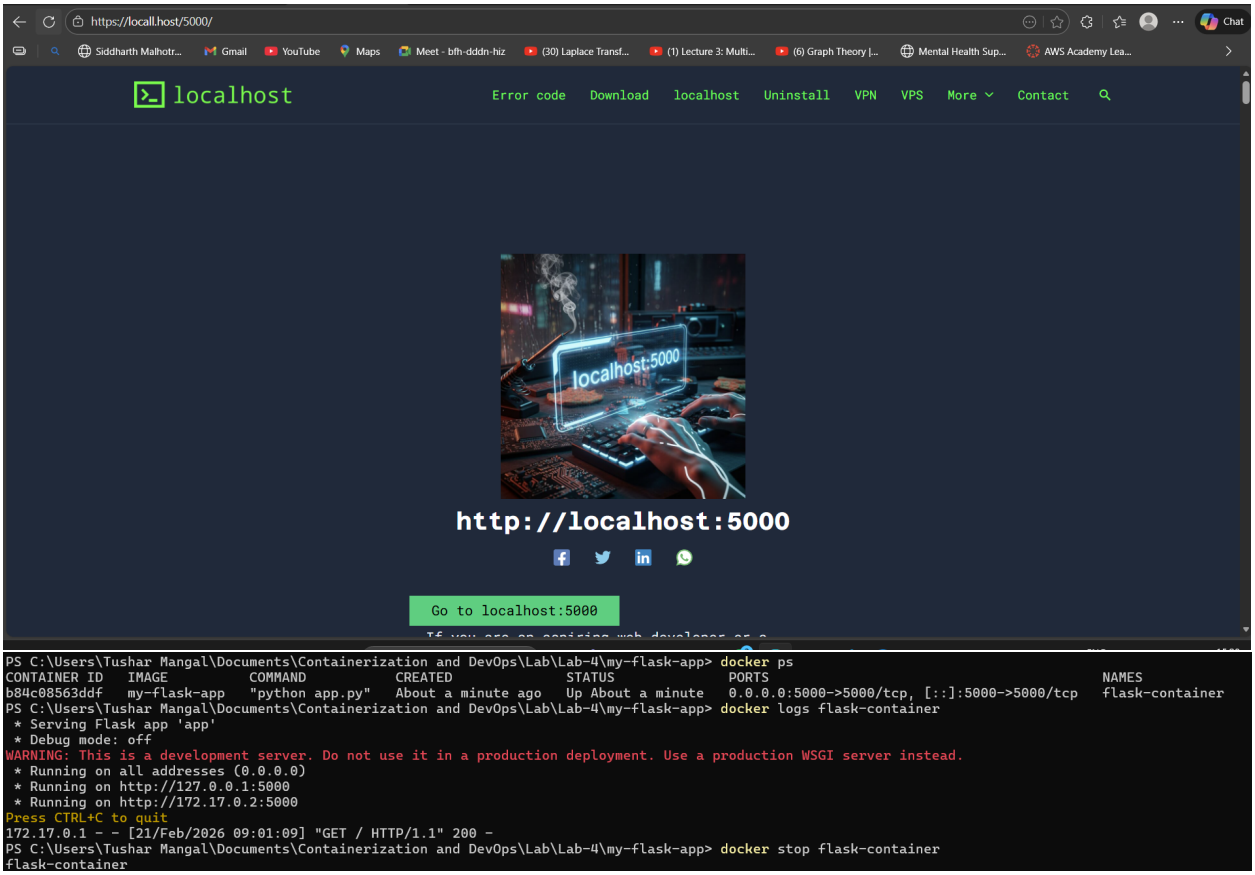
```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker images
                                                  i Info →   U  In Use
IMAGE                          ID             DISK USAGE   CONTENT SIZE   EXTRA
cloud-robot:latest             1ae3bf81d146       857MB         191MB    U
my-flask-app:1.0               67ab170e1c73       200MB        48.8MB
my-flask-app:latest            67ab170e1c73       200MB        48.8MB
my-flask-app:v1.0              67ab170e1c73       200MB        48.8MB
nginx-alpine:latest            3a47dbf1b178        16MB        4.74MB    U
nginx-ubuntu:latest            cb305251001b       199MB        52.7MB    U
nginx:latest                   341bf0f3ce6c       240MB        65.8MB    U
username/my-flask-app:1.0      678a8667bb82       200MB        48.8MB
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker history my-flask-app
IMAGE          CREATED              CREATED BY                                      SIZE      COMMENT
67ab170e1c73   About a minute ago   CMD ["python" "app.py"]                         0B        buildkit.dockerfile.v0
<missing>      About a minute ago   EXPOSE [5000/tcp]                               0B        buildkit.dockerfile.v0
<missing>      About a minute ago   COPY app.py . # buildkit                        12.3kB    buildkit.dockerfile.v0
<missing>      About a minute ago   RUN /bin/sh -c pip install --no-cache-dir -r…   13.4MB    buildkit.dockerfile.v0
<missing>      About a minute ago   COPY requirements.txt . # buildkit              12.3kB    buildkit.dockerfile.v0
<missing>      28 hours ago         WORKDIR /app                                    8.19kB    buildkit.dockerfile.v0
<missing>      3 months ago         CMD ["python3"]                                 0B        buildkit.dockerfile.v0
<missing>      3 months ago         RUN /bin/sh -c set -eux;  for src in idle3 p…   16.4kB    buildkit.dockerfile.v0
<missing>      3 months ago         RUN /bin/sh -c set -eux;   savedAptMark="$(a…   45.7MB    buildkit.dockerfile.v0
<missing>      3 months ago         ENV PYTHON_SHA256=00e07d7c0f2f0cc002432d1ee8…   0B        buildkit.dockerfile.v0
<missing>      3 months ago         ENV PYTHON_VERSION=3.9.25                       0B        buildkit.dockerfile.v0
<missing>      3 months ago         ENV GPG_KEY=E3FF2839C048B25C084DEBE9B26995E3…   0B        buildkit.dockerfile.v0
<missing>      3 months ago         RUN /bin/sh -c set -eux;  apt-get update;  a…   4.94MB    buildkit.dockerfile.v0
<missing>      3 months ago         ENV LANG=C.UTF-8                                0B        buildkit.dockerfile.v0
<missing>      3 months ago         ENV PATH=/usr/local/bin:/usr/local/sbin:/usr…   0B        buildkit.dockerfile.v0
<missing>      4 months ago         # debian.sh --arch 'amd64' out/ 'trixie' '@1…   87.4MB    debuerreotype 0.16
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker inspect my-flask-app
[
    {
        "Id": "sha256:67ab170e1c73b28dd6107a293781d17f87e045df9bf0683f1a5b4b74efd7d487",
        "RepoTags": [
            "my-flask-app:1.0",
            "my-flask-app:latest",
            "my-flask-app:v1.0"
        ],
        "RepoDigests": [
            "my-flask-app@sha256:67ab170e1c73b28dd6107a293781d17f87e045df9bf0683f1a5b4b74efd7d487"
        ],
```

10.5 Part 4: Running Containers

10.5.1 Step 1: Run Container

```
# Run container with port mapping
docker run -d -p 5000:5000 --name flask-container my-flask-app

# Test the application
curl http://localhost:5000

# View running containers
docker ps

# View container logs
docker logs flask-container
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker run -d -p 5000:5000 --name flask-container my-flask-app
b84c08563ddf12d17bb08a7bb460be1005f6f8646457177bcccb9cb32424ade8b
```



10.5.2 Step 2: Manage Containers

```
# Stop container
docker stop flask-container

# Start stopped container
docker start flask-container

# Remove container
docker rm flask-container

# Remove container forcefully
docker rm -f flask-container
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker stop flask-container
flask-container
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker start flask-container
flask-container
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker rm -f flask-container
flask-container
```

10.6 Part 5: Multi-stage Builds

10.6.1 Step 1: Why Multi-stage Builds?

- Smaller final image size
- Better security (remove build tools)
- Separate build and runtime environments

10.6.2 Step 2: Simple Multi-stage Dockerfile

Dockerfile.multistage:

```
# STAGE 1: Builder stage
FROM python:3.9-slim AS builder

WORKDIR /app

# Copy requirements
COPY requirements.txt .

# Install dependencies in virtual environment
RUN python -m venv /opt/venv
ENV PATH="/opt/venv/bin:$PATH"
RUN pip install --no-cache-dir -r requirements.txt

# STAGE 2: Runtime stage
FROM python:3.9-slim

WORKDIR /app

# Copy virtual environment from builder
COPY --from=builder /opt/venv /opt/venv
ENV PATH="/opt/venv/bin:$PATH"

# Copy application code
COPY app.py .

# Create non-root user
RUN useradd -m -u 1000 appuser
USER appuser

# Expose port
EXPOSE 5000

# Run application
CMD ["python", "app.py"]
```

Dockerfile.multistage

```
WORKDIR /app

# Copy requirements
COPY requirements.txt .

# Install dependencies in virtual environment
RUN python -m venv /opt/venv
ENV PATH="/opt/venv/bin:$PATH"
RUN pip install --no-cache-dir -r requirements.txt

# STAGE 2: Runtime stage
FROM python:3.9-slim

WORKDIR /app

# Copy virtual environment from builder
COPY --from=builder /opt/venv /opt/venv
ENV PATH="/opt/venv/bin:$PATH"

# Copy application code
COPY app.py .

# Create non-root user
RUN useradd -m -u 1000 appuser
USER appuser

# Expose port
EXPOSE 5000

# Run application
CMD ["python", "app.py"]
```

10.6.3 Step 3: Build and Compare

```
# Build regular image
docker build -t flask-regular .

# Build multi-stage image
docker build -f Dockerfile.multistage -t flask-multistage .

# Compare sizes
docker images | findstr flask
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker build -t flask-regular .
[+] Building 2.6s (10/10) FINISHED                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                              0.0s
 => => transferring dockerfile: 381B                                                                                              0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim                                                                2.0s
 => [internal] load .dockerignore                                                                                                 0.0s
 => => transferring context: 304B                                                                                                 0.0s
 => [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b          0.1s
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b          0.1s
 => [internal] load build context                                                                                                 0.0s
 => => transferring context: 63B                                                                                                  0.0s
 => CACHED [2/5] WORKDIR /app                                                                                                     0.0s
 => CACHED [3/5] COPY requirements.txt .                                                                                          0.0s
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                               0.0s
 => CACHED [5/5] COPY app.py .                                                                                                    0.0s
 => exporting to image                                                                                                            0.3s
 => => exporting layers                                                                                                           0.0s
 => => exporting manifest sha256:c8d2616072f3cdd22c334d17834f3d624565fc106dd2515036cda294b8739873                                 0.0s
 => => exporting config sha256:47bcef556a96a512e2e965ef7e584562a32fe168e094fe5ebbfce2e07da8904a                                   0.0s
 => => exporting attestation manifest sha256:4cdcf15979099a442cb0b7bad07431d034812899c7d35b501f5538ba79736b3f                     0.1s
 => => exporting manifest list sha256:b872d11baae281f2eac92ab62e961f50b542b29182d215f6f0c6beddce84928f                            0.0s
 => => naming to docker.io/library/flask-regular:latest                                                                           0.0s
 => => unpacking to docker.io/library/flask-regular:latest                                                                        0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/lm3lgckblt4ynxw9l6qweenc3
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker build -f Dockerfile.multistage -t flask-multistage .
[+] Building 31.3s (13/13) FINISHED                                                                                 docker:desktop-linux
 => [internal] load build definition from Dockerfile.multistage                                                                   0.0s
 => => transferring dockerfile: 709B                                                                                              0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim                                                                0.6s
 => [internal] load .dockerignore                                                                                                 0.0s
 => => transferring context: 304B                                                                                                 0.0s
 => [builder 1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b  0.1s
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b          0.1s
 => [internal] load build context                                                                                                 0.0s
 => => transferring context: 63B                                                                                                  0.0s
 => CACHED [builder 2/5] WORKDIR /app                                                                                             0.0s
 => CACHED [builder 3/5] COPY requirements.txt .                                                                                  0.0s
 => [builder 4/5] RUN python -m venv /opt/venv                                                                                   11.5s
 => [builder 5/5] RUN pip install --no-cache-dir -r requirements.txt                                                             10.8s
 => [stage-1 3/5] COPY --from=builder /opt/venv /opt/venv                                                                         1.5s
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker images | findstr flask
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
flask-multistage:latest     633a0182f68d        219MB         52.1MB
flask-regular:latest        b872d11baae2        200MB         48.8MB
my-flask-app:1.0            67ab170e1c73        200MB         48.8MB
my-flask-app:latest         67ab170e1c73        200MB         48.8MB
my-flask-app:v1.0           67ab170e1c73        200MB         48.8MB
username/my-flask-app:1.0   678a8667bb82        200MB         48.8MB
```

10.7 Part 6: Publishing to Docker Hub

10.7.1 Step 1: Prepare for Publishing

```
# Login to Docker Hub
docker login

# Tag image for Docker Hub
docker tag my-flask-app:latest username/my-flask-app:1.0
docker tag my-flask-app:latest username/my-flask-app:latest

# Push to Docker Hub
docker push username/my-flask-app:1.0
docker push username/my-flask-app:latest
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker login
Authenticating with existing credentials... [Username: userna1]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker tag my-flask-app:latest userna1/my-flask-app:1.0
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker tag my-flask-app userna1/my-flask-app:latest
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker push userna1/my-flask-app:1.0
The push refers to repository [docker.io/userna1/my-flask-app]
cabc5391baad: Already exists
fc7443084902: Layer already exists
b94d5f0ea6f3: Layer already exists
16b6cba43be5: Layer already exists
079b801f22d6: Layer already exists
ff6d87c41c49: Layer already exists
38513bd72563: Layer already exists
ea56f685404a: Layer already exists
b3ec39b36ae8: Layer already exists
1.0: digest: sha256:67ab170e1c73b28dd6107a293781d17f87e045df9bf0683f1a5b4b74efd7d487 size: 856
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker push userna1/my-flask-app:latest
The push refers to repository [docker.io/userna1/my-flask-app]
ff6d87c41c49: Pushed
fc7443084902: Pushed
38513bd72563: Pushed
b94d5f0ea6f3: Pushed
16b6cba43be5: Pushed
cabc5391baad: Pushed
b3ec39b36ae8: Pushed
ea56f685404a: Pushed
079b801f22d6: Pushed
latest: digest: sha256:67ab170e1c73b28dd6107a293781d17f87e045df9bf0683f1a5b4b74efd7d487 size: 856
```

10.7.2 Step 2: Pull and Run from Docker Hub

```
# Pull from Docker Hub (on another machine)
docker pull username/my-flask-app:latest

# Run the pulled image
docker run -d -p 5000:5000 username/my-flask-app:latest
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker pull userna1/my-flask-app:latest
latest: Pulling from userna1/my-flask-app
Digest: sha256:67ab170e1c73b28dd6107a293781d17f87e045df9bf0683f1a5b4b74efd7d487
Status: Image is up to date for userna1/my-flask-app:latest
docker.io/userna1/my-flask-app:latest
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4\my-flask-app> docker run -d -p 5000:5000 userna1/my-flask-app:latest
53270aa75836bd0b71b87a62e807419d730b4c6bcb6bd4f5e7f3b4357d449655
```

10.8 Part 7: Node.js Example (Quick Version)

10.8.1 Step 1: Node.js Application

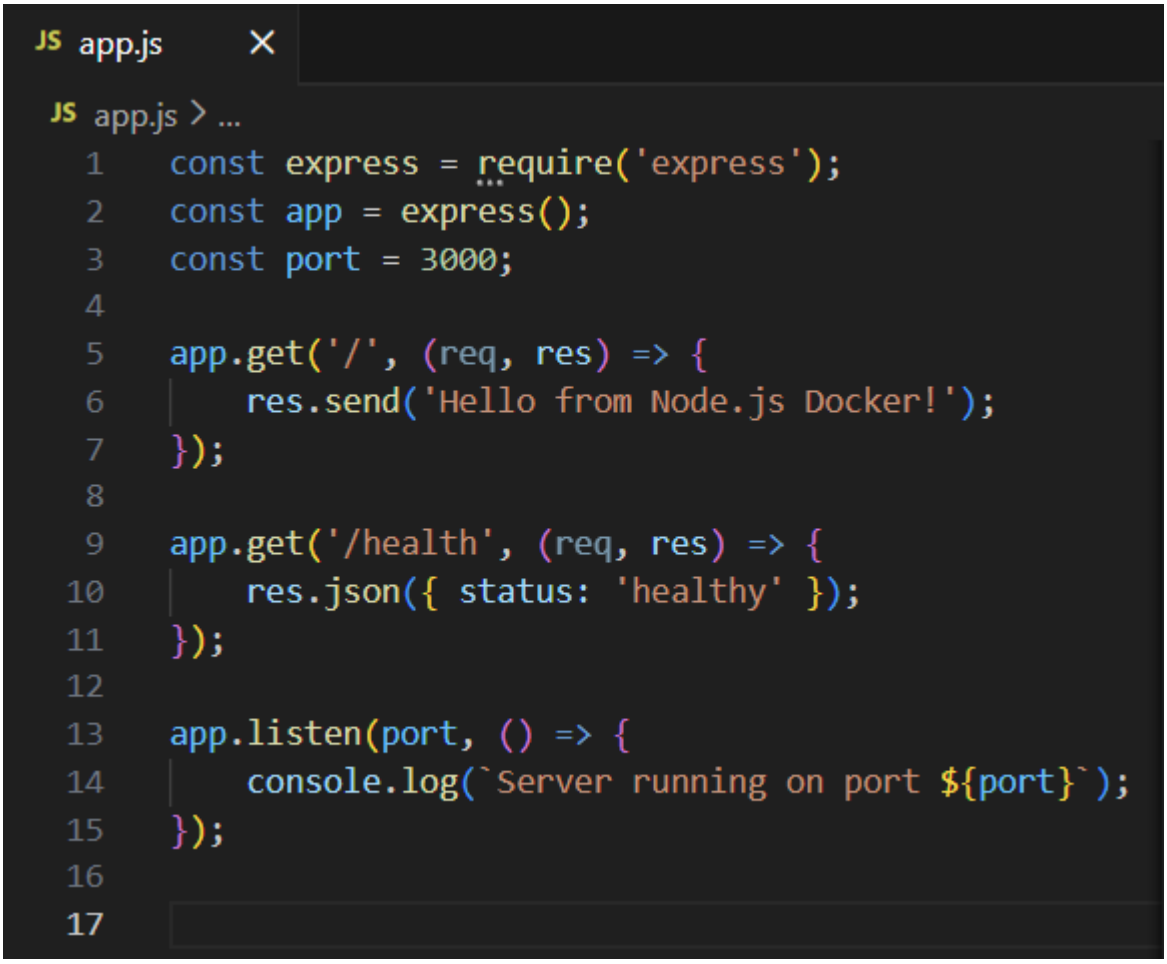
```
mkdir my-node-app
cd my-node-app
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4> mkdir my-node-app


    Directory: C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        21-02-2026     15:04                my-node-app


PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-4> cd my-node-app
```

app.js:

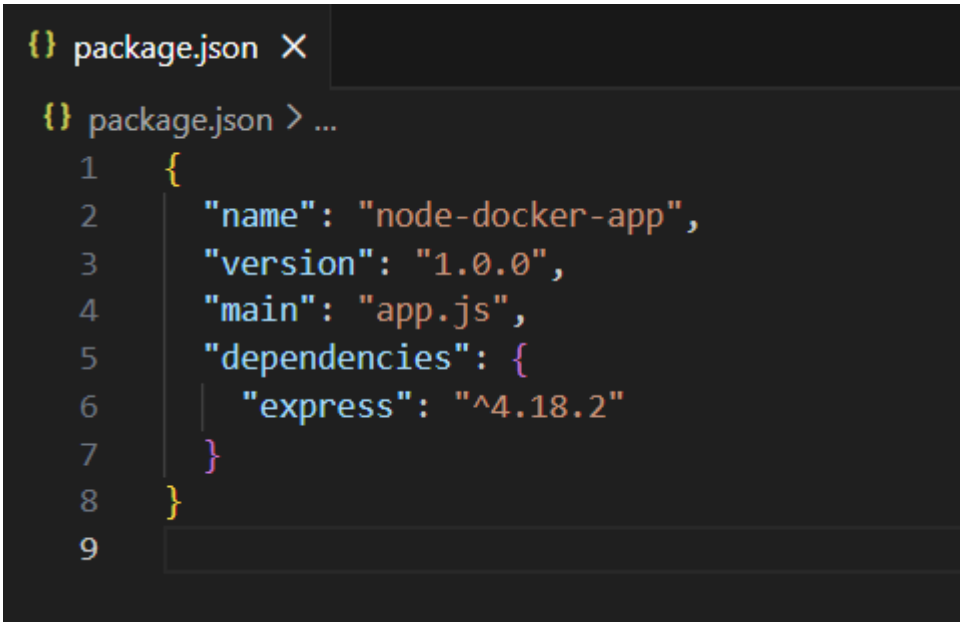
```
const express = require('express');
const app = express();
const port = 3000;

app.get('/', (req, res) => {
```

```
    res.send('Hello from Node.js Docker!');
});

app.get('/health', (req, res) => {
    res.json({ status: 'healthy' });
});

app.listen(port, () => {
    console.log(`Server running on port ${port}`);
});
```



package.json:

```
{
  "name": "node-docker-app",
  "version": "1.0.0",
  "main": "app.js",
  "dependencies": {
    "express": "^4.18.2"
  }
}
```



10.8.2 Step 2: Node.js Dockerfile

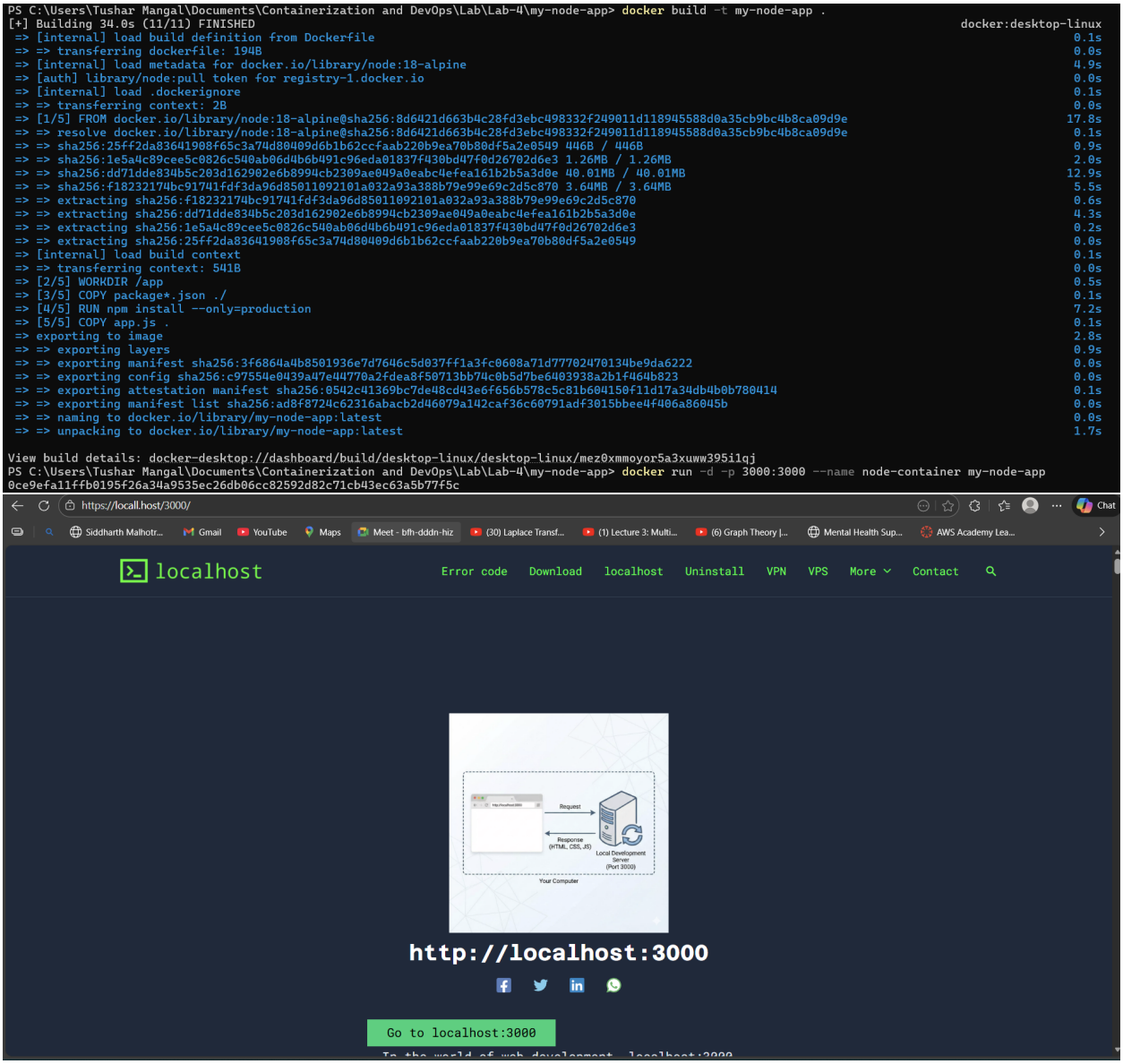
```
FROM node:18-alpine

WORKDIR /app

COPY package*.json ./
RUN npm install --only=production

COPY app.js .

EXPOSE 3000

CMD ["node", "app.js"]
```

Dockerfile

```
1   FROM node:18-alpine
2
3   WORKDIR /app
4
5   COPY package*.json ./
6   RUN npm install --only=production
7
8   COPY app.js .
9
10  EXPOSE 3000
11
12  CMD ["node", "app.js"]
13
```

10.8.3 Step 3: Build and Run

```
# Build image
docker build -t my-node-app .

# Run container
docker run -d -p 3000:3000 --name node-container my-node-app

# Test
curl http://localhost:3000
```



11 Containerization-and-DevOps

12 Lab 5: Docker Volumes, Environment Variables, Monitoring & Networks

12.1 Objective

Learn advanced Docker concepts including persistent data storage with volumes, application configuration with environment variables, container monitoring, and linking multiple containers using Docker networks.

12.2 Part 1: Docker Volumes – Persistent Data Storage

By default, data inside a container is ephemeral (lost when the container is removed). Volumes provide a way to safely persist data.

12.2.1 1. Anonymous & Named Volumes

```
docker run -d -v /app/data --name web1 nginx
docker volume create mydata
docker run -d -v mydata:/app/data --name web2 nginx
docker volume ls
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5> docker run -d -v /app/data --name web1 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
448ea5cac5d5: Pull complete
5435b2dcdf5c: Pull complete
7b5d674621c2: Pull complete
84e114c2bb36: Pull complete
054715a6bffa: Pull complete
4a038fd18db1: Pull complete
88d1d984b765: Pull complete
3eff0a97d435: Download complete
cc0cf959117b: Download complete
Digest: sha256:7f0adca1fc6c29c8dc49a2e90037a10ba20dc266baaed0988e9fb4d0d8b85ba0
Status: Downloaded newer image for nginx:latest
c0f2f3083352ae9f59c351f1f4bc844e6ceaed355d383acc36e6c49623dba100
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5> docker volume create mydata
mydata
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5> docker run -d -v mydata:/app/data --name web2 nginx
db76aaf4574ffadb6accacbe42547494d339608cac2a03bb13705561e0905b57
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5> docker volume ls
DRIVER    VOLUME NAME
local     78e3658005c9633c191021bfb9bf73c4706f897e0900622451e3c1e3eea2845a
local     assingment1_postgres_data
local     assingment-1_pgdata
local     mydata
```

12.2.2 2. Bind Mounts (Host Directory)

Bind mounts attach a specific directory on your host machine to a directory inside the container.

```
mkdir myapp-data
docker run -d -v ${PWD}/myapp-data:/app/data --name web3 nginx
echo "From Host" > myapp-data/Text-file.txt
docker exec web3 cat /app/data/Text-file.txt
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5> mkdir myapp-data


    Directory: C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        09-04-2026     10:52                myapp-data


PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5> docker run -d -v ${PWD}/myapp-data:/app/data --name web3 nginx
d325e4eb7b76c7f289a04398f122732ad516a77d592a57909e0092d807b92d94
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5> echo "The Text File" > myapp-data/Text-file.txt
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5> docker exec web3 cat /app/data/Text-file.txt
◆◆The Text File
```

12.3 Part 2: Environment Variables

Environment variables allow us to pass configuration parameters into the container at runtime.

12.3.1 1. Building and Running a Flask App with ENV Vars

We built a custom Flask application that returns configuration data via a `/config` API endpoint.

app.py

```
import os
from flask import Flask
app = Flask(__name__)

db_host = os.environ.get('DATABASE_HOST', 'localhost')
debug_mode = os.environ.get('DEBUG', 'false').lower() == 'true'
api_key = os.environ.get('API_KEY')

@app.route('/config')
def config():
    return {
        'db_host': db_host,
        'debug': debug_mode,
        'has_api_key': bool(api_key)
    }

if __name__ == '__main__':
    port = int(os.environ.get('PORT', 5000))
    app.run(host='0.0.0.0', port=port, debug=debug_mode)
```

Dockerfile

```
FROM python:3.9-slim

ENV PYTHONUNBUFFERED=1
ENV PYTHONDONTWRITEBYTECODE=1
```

```
WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt

COPY app.py .

ENV PORT=5000
ENV DEBUG=false

EXPOSE 5000
CMD ["python", "app.py"]
```

Lab > Lab-5 > flask-env > Dockerfile > FROM

```
1   FROM python:3.9-slim
2
3   ENV PYTHONUNBUFFERED=1
4   ENV PYTHONDONTWRITEBYTECODE=1
5
6   WORKDIR /app
7   COPY requirements.txt .
8   RUN pip install -r requirements.txt
9
10  COPY app.py .
11
12  ENV PORT=5000
13  ENV DEBUG=false
14
15  EXPOSE 5000
16  CMD ["python", "app.py"]
```

```
docker build -t flask-env-app .
docker run -d --name flask-app -p 5000:5000 \
  -e DATABASE_HOST="prod-db.example.com" \
  -e DEBUG="true" \
  -e API_KEY="secret-xyz" \
  flask-env-app
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5\flask-env> docker build -t flask-env .
[+] Building 39.0s (10/10) FINISHED                                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                  0.0s
 => => transferring dockerfile: 286B                                                                                  0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim                                                    2.1s
 => [internal] load .dockerignore                                                                                     0.0s
 => => transferring context: 2B                                                                                       0.0s
 => [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13c     19.8s
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf      0.1s
 => => sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 13.88MB / 13.88MB                     15.0s
 => => sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 251B / 251B                            2.2s
 => => sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 1.29MB / 1.29MB                        3.8s
 => => sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 29.78MB / 29.78MB                     12.7s
 => => extracting sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d                             3.9s
 => => extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d                             0.4s
 => => extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08                             2.4s
 => => extracting sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508                             0.0s
 => [internal] load build context                                                                                     0.0s
 => => transferring context: 85B                                                                                      0.0s
 => [2/5] WORKDIR /app                                                                                                0.7s
 => [3/5] COPY requirements.txt .                                                                                     0.1s
 => [4/5] RUN pip install -r requirements.txt                                                                        13.9s
 => [5/5] COPY app.py .                                                                                               0.1s
 => exporting to image                                                                                                1.8s
 => => exporting layers                                                                                               0.9s
 => => exporting manifest sha256:d3666df891d65fe6aaa60c715a2aa8df7d8b0e64430bb3dafe60a7f78aa17032                      0.0s
 => => exporting config sha256:d012cfcf51a1c37372e6aa95d88a2617715e21653b474316df588dffbb530356                       0.0s
 => => exporting attestation manifest sha256:6a463955c51f27a3af55830005c442a4e9f1d4ccf9473b29f435c392f3cb6ebd         0.1s
 => => exporting manifest list sha256:2675e06f84047fcb1af42e562e6f857942636b9f5bb8e0ccd7346961c134c354                0.0s
 => => naming to docker.io/library/flask-env:latest                                                                   0.0s
 => => unpacking to docker.io/library/flask-env:latest                                                                0.7s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/5agu5ypbn08rh4pwct9bi5p5i
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5\flask-env> docker run -d --name flask-app -p 5000:5000 \
docker: invalid reference format

Run 'docker run --help' for more information
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5\flask-env> docker run -d --name flask-app -p 5000:5000 -e DATABASE_HOST="prod-db.e
xample.com" -e DEBUG="true" -e API_KEY="secret-xyz" flask-env
0e2d21bb3ddeb2a19fa03b272d05a68f7370ac550f875cdefa51ff7fc38a9d22
```

12.3.2 2. Testing Injected Environment Variables

We can verify that the environment variables were successfully passed into the running container.

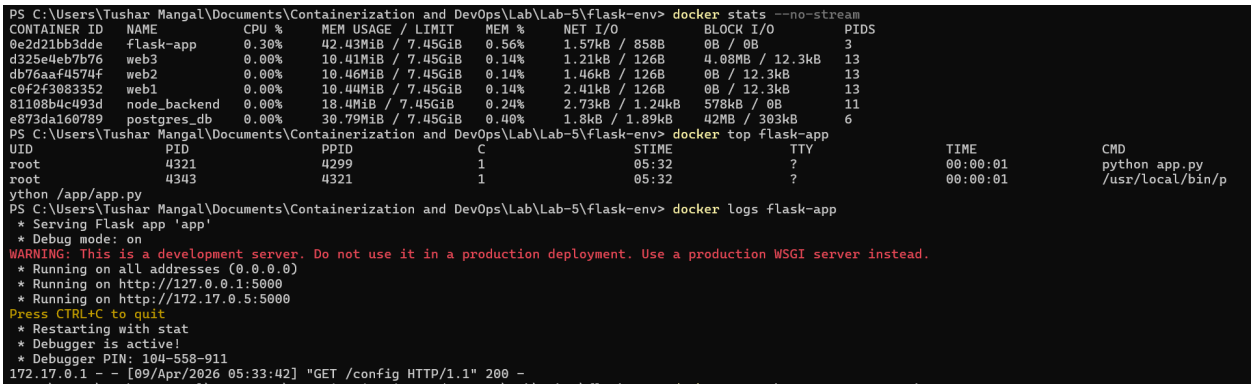
```
docker exec flask-app printenv DATABASE_HOST
curl http://localhost:5000/config
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5\flask-env> docker exec flask-app printenv DATABASE_HOST
prod-db.example.com
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5\flask-env> curl http://localhost:5000/config

Security Warning: Script Execution Risk
Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is parsed.
      RECOMMENDED ACTION:
      Use the -UseBasicParsing switch to avoid script code execution.

      Do you want to continue?

[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "N"): y


StatusCode        : 200
StatusDescription : OK
Content           : {
                      "db_host": "prod-db.example.com",
                      "debug": true,
                      "has_api_key": true
                    }

RawContent        : HTTP/1.1 200 OK
                    Connection: close
                    Content-Length: 79
                    Content-Type: application/json
                    Date: Thu, 09 Apr 2026 05:33:42 GMT
                    Server: Werkzeug/3.1.8 Python/3.9.25

                    {
                      "db_host": "prod-db.example.com...
Forms             : {}
Headers           : {[Connection, close], [Content-Length, 79], [Content-Type, application/json], [Date, Thu, 09 Apr 2026 05:33:42 GMT]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 79
```

12.4 Part 3: Docker Monitoring

Monitoring allows us to keep track of resource usage, running processes, and application logs.

```
docker stats --no-stream
docker top flask-app
docker logs flask-app
```



12.4.1 Automated Monitoring Script

We can automate these checks using a bash script (`monitor.sh`):

monitor.sh

```
#!/bin/bash

echo "=== Docker Monitoring Dashboard ==="
echo "Time: $(date)"

echo "1. Running Containers:"
docker ps --format "table \t\t"

echo "2. Resource Usage:"
docker stats --no-stream --format "table \t\t"

echo "3. Recent Events:"
docker events --since '5m' --until '0s' | tail

echo "4. System Info:"
docker system df
```

Lab > Lab-5 > $ monitor.sh

```
#!/bin/bash

echo "=== Docker Monitoring Dashboard ==="
echo "Time: $(date)"

echo "1. Running Containers:"
docker ps --format "table {{.Names}}\t{{.Status}}\t{{.Ports}}"

echo "2. Resource Usage:"
docker stats --no-stream --format "table {{.Name}}\t{{.CPUPerc}}\t{{.MemUsage}}"

echo "3. Recent Events:"
docker events --since '5m' --until '0s' | tail

echo "4. System Info:"
docker system df
```

12.5 Part 4: Docker Networks

Docker networks allow isolated containers to securely communicate with each other.

12.5.1 1. Creating a Network

We created a custom bridge network and attached two Nginx containers to it.

```
docker network create my-network
docker run -d --name net-web1 --network my-network nginx
docker run -d --name net-web2 --network my-network nginx
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5\flask-env> docker network create my-network
836ff7025198ea7a06e0df96b0fe8e06d3eaa73bbad44656e7ae43ce2a8c2a07
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5\flask-env> docker run -d --name net-web1 --network my-network nginx
620c6532ae622799ffe895275165aa8dcbf79945709f15690d41f01be25ea354
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5\flask-env> docker run -d --name net-web2 --network my-network nginx
ae9151683ab7c826983516761d5caa28695f0071191cddd89267e15798f64066
```

12.5.2 2. Internal Network Communication

Because they share a custom network, containers can communicate with each other using their container names as DNS addresses.

```
docker exec net-web1 curl -s http://net-web2
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5\flask-env> docker exec net-web1 curl -s http://net-web2
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

12.6 Part 5: Real-World Example (Multi-Container Setup)

In a real-world scenario, we often have an application connected to a database and a cache layer, all residing on the same network.

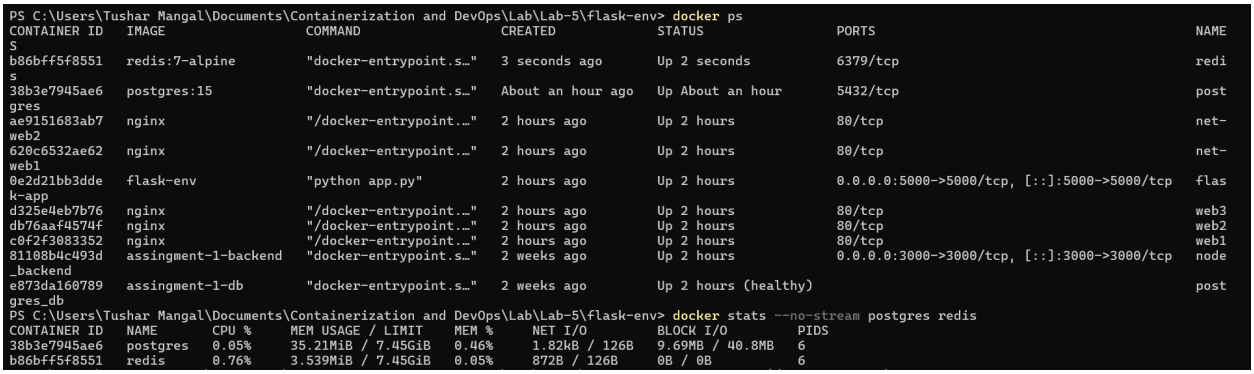
12.6.1 1. Running Postgres and Redis on a Shared Network

```
docker network create myapp-network
docker run -d --name postgres --network myapp-network \
  -e POSTGRES_PASSWORD=mysecretpassword \
  -v postgres-data:/var/lib/postgresql/data postgres:15

docker run -d --name redis --network myapp-network \
  -v redis-data:/data redis:7-alpine
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5\flask-env> docker network create myapp-network
481903d333ce683b9a1d1ca9fc1766673edbaa98c173e68efc8889d72dd0fa68
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5\flask-env> docker run -d --name postgres --network myapp-network -e POSTGRES_PASSW
ORD=mysecretpassword -v postgres-data:/var/lib/postgresql/data postgres:15
Unable to find image 'postgres:15' locally
15: Pulling from library/postgres
71a32c7571f4: Pull complete
a8a0725a32a6: Pull complete
5c152148754e: Pull complete
b4765ac2e7e6: Pull complete
af2a6e5f285c: Pull complete
5ca81a59b0a4: Pull complete
00f33d580dc3: Pull complete
8ecdbd190da6: Pull complete
c45b346571fc: Pull complete
2c38b75ffca2: Pull complete
af0c3ea2fe5e: Pull complete
9328c154beae: Pull complete
f7049a93e93f: Pull complete
cebcba07837b: Download complete
b9015a4f6b87: Download complete
Digest: sha256:31224c60946ea75ee9522e0f4d02eb202e74754adea4294fdefe9f93c875b657
Status: Downloaded newer image for postgres:15
38b3e7945ae62962194fce9f6df511c8295115038146aa414308615d78c338e2
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-5\flask-env> docker run -d --name redis --network myapp-network -v redis-data:/data
redis:7-alpine
Unable to find image 'redis:7-alpine' locally
7-alpine: Pulling from library/redis
1c2c8d1ee428: Pull complete
09a5a0c32a23: Pull complete
7ad54b3c4cef: Pull complete
4f4fb700ef54: Pull complete
9fe3744a2eac: Pull complete
2a97533d89ca: Pull complete
bd53938b1271: Pull complete
bc1da058f299: Pull complete
9bbcfa5d5580: Download complete
06cd2b88db40: Download complete
Digest: sha256:8b81dd37ff027bec4e516d41acfbe9fe2460070dc6d4a4570a2ac5b9d59df065
Status: Downloaded newer image for redis:7-alpine
b86bff5f8551a000e5fb997aa1caca301215742bd33e172d6fc6c954ad5a2cfb
```

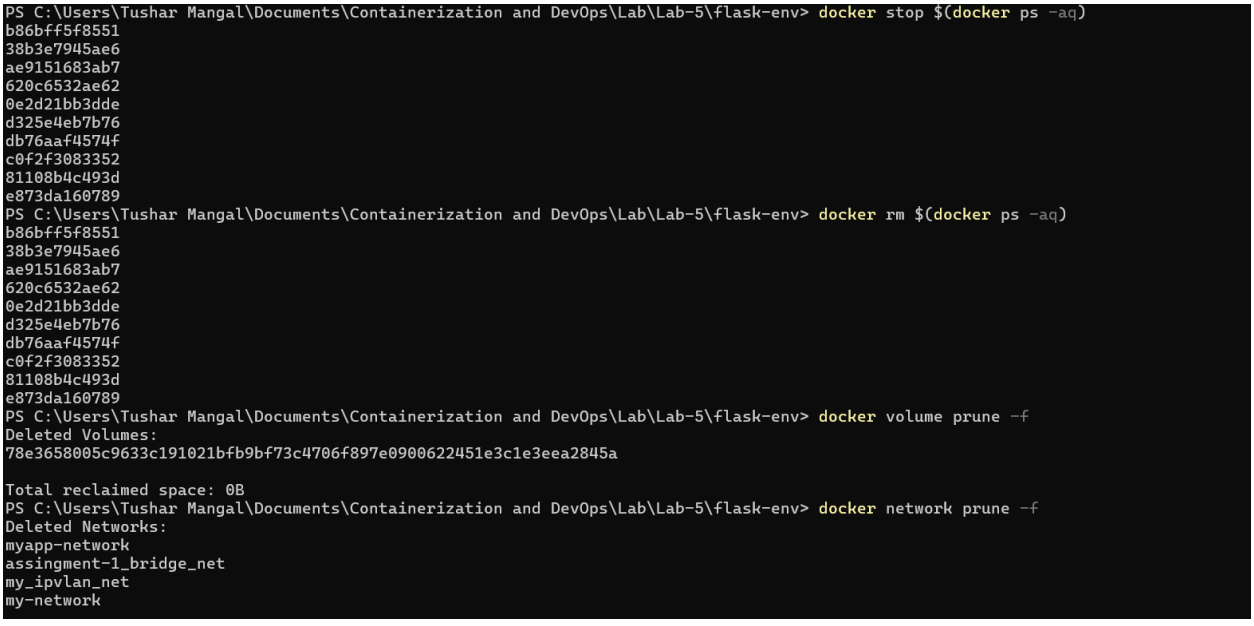
12.6.2 2. Verifying the Services

```
docker ps
docker stats --no-stream postgres redis
```



12.7 Cleanup

```
docker stop $(docker ps -aq)
docker rm $(docker ps -aq)
docker volume prune -f
docker network prune -f
```



12.8 Docker Commands Cheat Sheet

12.8.1 Volumes

| Command | Description | |———|————-| | `docker volume create <name>` | Create a named volume | | `docker run -v <volume>:/path` | Mount named volume | | `docker run -v /host:/container` | Mount local bind |

12.8.2 Environment Variables

| Command | Description | |———|————-| | `docker run -e VAR=value` | Inject single variable | | `docker run --env-file .env` | Inject multiple from file |

12.8.3 Monitoring

| Command | Description | |———|————-| | `docker stats` | Live resource usage | | `docker top <container>` | process list inside container | | `docker logs -f <container>` | Follow container logs |

12.8.4 Networks

| Command | Description | |———|————-| | `docker network create <name>` | Create custom network | | `docker run --network <name>` | Run container on network |

12.9 Key Takeaways

1. **Volumes persist data** safely outside the container lifecycle.
2. **Env variables configure apps** without modifying the actual code or image.
3. **Monitoring helps debugging** performance, memory issues, and hidden errors.
4. **Networks enable communication** securely via DNS isolation between containers.
5. Use **named volumes in production** for database persistence.
6. Use **custom networks** for multi-container applications instead of linking containers manually.

---

13 Containerization-and-DevOps

14 Experiment 6A: Comparison of Docker Run and Docker Compose

14.1 PART A – THEORY

14.1.1 1. Objective

To understand the relationship between Docker Run and Docker Compose, and compare their syntax and use cases.

14.1.2 2. Background Theory

14.1.2.1 2.1 Docker Run (Imperative Approach)

- Used to create and start containers.
- Requires manual flags for everything (Ports, Volumes, Networks, Restart policies, Resources, Names).

Example:

```
docker run -d \
 --name my-nginx \
 -p 8080:80 \
 -v ./html:/usr/share/nginx/html \
 -e NGINX_HOST=localhost \
 --restart unless-stopped \
 nginx:alpine
```

14.1.2.2 2.2 Docker Compose (Declarative Approach)

- Uses `docker-compose.yml` to define services, networks, and volumes in a single file.
- One command (`docker compose up -d`) runs everything.

Equivalent Compose File:

```
version: '3.8'
services:
  nginx:
    image: nginx:alpine
    container_name: my-nginx
    ports:
      - "8080:80"
    volumes:
      - ./html:/usr/share/nginx/html
    environment:
      NGINX_HOST: localhost
    restart: unless-stopped
```

14.1.3 3. Mapping: Docker Run vs Compose

Docker Run	Docker Compose
`-p`	`ports`
`-v`	`volumes`
`-e`	`environment`
`--name`	`container_name`
`--network`	`networks`
`--restart`	`restart`
`--memory`	`deploy.resources.limits.memory`
`--cpus`	`deploy.resources.limits.cpus`

14.1.4 4. Advantages of Docker Compose

- Multi-container support
- Reproducibility & Version control
- Unified lifecycle commands
- Scaling capability (`docker compose up --scale web=3`)

14.2 PART B – PRACTICAL TASKS

14.2.1 Task 1: Single Container Comparison

14.2.1.1 Using Docker Run

```
docker run -d --name lab-nginx -p 8081:80 -v $(pwd)/html:/usr/share/nginx/html nginx:alpine
curl http://localhost:8081
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-6\task1-single> echo "Hello docker"> html/index.html
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-6\task1-single> docker run -d --name lab-nginx -p 8081:80 -v ${PWD}/html:/usr/share/
nginx/html nginx:alpine
Unable to find image 'nginx:alpine' locally
alpine: Pulling from library/nginx
ff9f59a6a62e: Pull complete
1165b869c51a: Pull complete
a71873b303e8: Pull complete
15e759724ff6: Pull complete
34dfdd2ef1f9: Pull complete
f03becc8ac15: Pull complete
c8a2fa3a88d2: Pull complete
6057002906d9: Download complete
1116431dccff: Download complete
Digest: sha256:645eda1c2477aaa9b879f73909b9222c6f19798dd45be6706268d82a661c6e6d
Status: Downloaded newer image for nginx:alpine
6218ed65efd04d9332729c3b421ed18f0f2846d50dd46a923991ab5cabe49667
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-6\task1-single> curl http://localhost:8081

Security Warning: Script Execution Risk
Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is parsed.
      RECOMMENDED ACTION:
      Use the -UseBasicParsing switch to avoid script code execution.

      Do you want to continue?

[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "N"): y


StatusCode        : 200
StatusDescription : OK
Content           : ÿþHello docker

RawContent        : HTTP/1.1 200 OK
                    Connection: keep-alive
                    Accept-Ranges: bytes
                    Content-Length: 30
                    Content-Type: text/html
                    Date: Thu, 09 Apr 2026 08:57:37 GMT
                    ETag: "69d769e5-1e"
                    Last-Modified: Thu, 09 Apr 2026 08...
                    Last-Modified: Thu, 09 Apr 2026 08...
Forms             : {}
Headers           : {[Connection, keep-alive], [Accept-Ranges, bytes], [Content-Length, 30], [Content-Type, text/html]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 30
```

14.2.1.2 Using Docker Compose docker-compose.yml

```
version: '3.8'
services:
  nginx:
    image: nginx:alpine
    container_name: lab-nginx
    ports:
      - "8081:80"
    volumes:
      - ./html:/usr/share/nginx/html
```

```
docker compose up -d
docker compose ps
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-6\task1-single> docker compose up -d
time="2026-04-09T14:49:02+05:30" level=warning msg="C:\\Users\\Tushar Mangal\\Documents\\Containerization and DevOps\\Lab\\Lab-6\\task1-single\\docker-compo
se.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 2/2
 ✔Network task1-single_default Created                                                                                                    0.1s
 ✔Container lab-nginx          Created                                                                                                    0.2s
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-6\task1-single> curl http://localhost:8081

Security Warning: Script Execution Risk
Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is parsed.
      RECOMMENDED ACTION:
      Use the -UseBasicParsing switch to avoid script code execution.

      Do you want to continue?

[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "N"): y


StatusCode        : 200
StatusDescription : OK
Content           : ÿþHello docker

RawContent        : HTTP/1.1 200 OK
                    Connection: keep-alive
                    Accept-Ranges: bytes
                    Content-Length: 30
                    Content-Type: text/html
                    Date: Thu, 09 Apr 2026 09:19:25 GMT
                    ETag: "69d769e5-1e"
                    Last-Modified: Thu, 09 Apr 2026 08...
Forms             : {}
Headers           : {[Connection, keep-alive], [Accept-Ranges, bytes], [Content-Length, 30], [Content-Type, text/html]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 30
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-6\task1-single> docker compose ps
time="2026-04-09T14:49:45+05:30" level=warning msg="C:\\Users\\Tushar Mangal\\Documents\\Containerization and DevOps\\Lab\\Lab-6\\task1-single\\docker-compo
se.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME        IMAGE          COMMAND                  SERVICE   CREATED          STATUS          PORTS
lab-nginx   nginx:alpine   "/docker-entrypoint.…"   nginx     42 seconds ago   Up 41 seconds   0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
```

14.2.2 Task 2: Multi-Container App (WordPress + MySQL)

14.2.2.1 Using Docker Run (Manual approach)

```
docker network create wp-net
```

```
docker run -d --name mysql --network wp-net -e MYSQL_ROOT_PASSWORD=secret -e MYSQL_DATABASE=wordpress my
```

```
docker run -d --name wordpress --network wp-net -p 8082:80 -e WORDPRESS_DB_HOST=mysql -e WORDPRESS_DB_PA
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-6\task1-single> docker network create wp-net
ea6c8498d440ec224dcad6f9ad7fbf1303e24b5b34493fab599926a40f385a4c
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-6\task1-single> docker run -d --name mysql --network wp-net -e MYSQL_ROOT_PASSWORD=s
ecret -e MYSQL_DATABASE=wordpress mysql:5.7
Unable to find image 'mysql:5.7' locally
5.7: Pulling from library/mysql
df9a4d85569b: Pull complete
68c3898c2015: Pull complete
20e4dcae4c69: Pull complete
ffc89e9dfd88: Pull complete
43d05e938198: Pull complete
6b95a940e7b6: Pull complete
ae71319cb779: Pull complete
90986bb8de6e: Pull complete
e9f03a1c24ce: Pull complete
1c56c3d4ce74: Pull complete
064b2d298fba: Pull complete
96889c47a44a: Download complete
9eb6a52b8441: Download complete
Digest: sha256:4bc6bc963e6d8443453676cae56536f4b8156d78bae03c0145cbe47c2aad73bb
Status: Downloaded newer image for mysql:5.7
1cb945e39eeb42f7ed8b2baa38c7ffa1614063e168d58cff14a0440d3234259f
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-6\task1-single> docker run -d --name wordpress --network wp-net -p 8082:80 -e WORDPR
ESS_DB_HOST=mysql -e WORDPRESS_DB_PASSWORD=secret wordpress:latest
Unable to find image 'wordpress:latest' locally
latest: Pulling from library/wordpress
e24121de25e6: Pull complete
90be0470639d: Pull complete
871c6a1a47cc: Pull complete
6562f7228ec3: Pull complete
1fa182c1d458: Pull complete
568912d7e28d: Pull complete
78a70974d738: Pull complete
4443d9c71a47: Pull complete
b4b1305df06a: Pull complete
d5af5dfa43d4: Pull complete
e052461f1b41: Pull complete
73ce216c5445: Pull complete
79e11af4b256: Pull complete
9bde0c34d1f4: Pull complete
271a7052ce75: Pull complete
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-6\task1-single> docker compose ps
time="2026-04-09T14:53:43+05:30" level=warning msg="C:\\Users\\Tushar Mangal\\Documents\\Containerization and DevOps\\Lab\\Lab-6\\task1-single\\docker-compo
se.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME        IMAGE          COMMAND                  SERVICE   CREATED          STATUS          PORTS
lab-nginx   nginx:alpine   "/docker-entrypoint.…"   nginx     4 minutes ago    Up 4 minutes    0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
```

(Note: The equivalent steps in Compose are demonstrated in Experiment 6B below).

14.3 PART C – CONVERSION TASKS (Theory)

Converting manual `docker run` commands and configurations into declarative `docker-compose.yml` configurations:

14.3.1 Task 3: Convert Run → Compose

Given Run Command:

```
docker run -d --name webapp -p 5000:5000 -e APP_ENV=production -e DEBUG=false --restart unless-stopped
```

Converted Compose File:

```
version: '3.8'
services:
  webapp:
    image: node:18-alpine
    container_name: webapp
    ports:
      - "5000:5000"
    environment:
      APP_ENV: production
      DEBUG: false
    restart: unless-stopped
```

14.3.2 Task 4 & 5: Volumes, Networks, and Resource Limits

Compose clearly organizes dependencies, limits, and custom named networks natively without extra CLI scaffolding. *(See lab theory notes for full limit syntax using `deploy:`).*

14.4 PART D – DOCKERFILE WITH COMPOSE

We can combine custom image building with Compose orchestration.

app.js

```
const http = require('http');
http.createServer((req, res) => {
  res.end("Docker Compose Build Lab");
}).listen(3000);
```

Dockerfile

```
FROM node:18-alpine
WORKDIR /app
COPY app.js .
EXPOSE 3000
CMD ["node", "app.js"]
```

docker-compose.yml

```
version: '3.8'
services:
  nodeapp:
```

```
    build:
      context: .
      dockerfile: Dockerfile
    container_name: custom-node-app
    ports:
      - "3000:3000"
```

```
docker compose up --build -d
curl http://localhost:3000
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-6\part-d-node> docker compose up --build -d
time="2026-04-09T19:44:15+05:30" level=warning msg="C:\\Users\\Tushar Mangal\\Documents\\Containerization and DevOps\\Lab\\Lab-6\\part-d-node\\docker-compos
e.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
#1 [internal] load local bake definitions
#1 reading from stdin 643B done
#1 DONE 0.0s

#2 [internal] load build definition from dockerfile
#2 transferring dockerfile: 122B 0.0s done
#2 DONE 0.1s

#3 [internal] load metadata for docker.io/library/node:18-alpine
#3 DONE 0.1s

#4 [internal] load .dockerignore
#4 transferring context:
#4 transferring context: 2B done
#4 DONE 0.1s

#5 [internal] load build context
#5 transferring context: 155B 0.0s done
#5 DONE 0.1s

#6 [1/3] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e
#6 resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e 0.1s done
#6 DONE 0.1s

#7 [2/3] WORKDIR /app
#7 CACHED

#8 [3/3] COPY app.js .
#8 DONE 0.1s

#9 exporting to image
#9 exporting layers
#9 exporting layers 0.2s done
#9 exporting manifest sha256:17bc1b78d4d95de13086f27f4092f59fe189716cfaa72d8c608bb8b7d61ef4d2 0.0s done
#9 exporting config sha256:acdb06cedbba30898e2f3b8bd3147f2c6a4c4715a8cbb0d2fa0bc6d007a57484 0.0s done
#9 exporting attestation manifest sha256:9faeb86f7a039aa3609c76db5a8c3b88f546738e0b11130690001147afa9ae6b
#9 exporting attestation manifest sha256:9faeb86f7a039aa3609c76db5a8c3b88f546738e0b11130690001147afa9ae6b 0.1s done
#9 exporting manifest list sha256:197c4bb2e0fda31ae76e00fff9e4603e2cd91d6a29366fc637d67cf7af03452a 0.0s done
#10 resolving provenance for metadata file
#10 DONE 0.0s
[+] up 3/3
 ✔Image part-d-node-nodeapp   Built                                                                                    2.5s
 ✔Network part-d-node_default Created                                                                                  0.1s
 ✔Container custom-node-app   Created                                                                                  0.2s
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-6\part-d-node> curl http://localhost:3000

Security Warning: Script Execution Risk
Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is parsed.
     RECOMMENDED ACTION:
     Use the -UseBasicParsing switch to avoid script code execution.

     Do you want to continue?

[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "N"): y


StatusCode        : 200
StatusDescription : OK
Content           : {68, 111, 99, 107...}
RawContent        : HTTP/1.1 200 OK
                    Connection: keep-alive
                    Keep-Alive: timeout=5
                    Content-Length: 24
                    Date: Thu, 09 Apr 2026 14:14:56 GMT

                    Docker Compose Build Lab
Headers           : {[Connection, keep-alive], [Keep-Alive, timeout=5], [Content-Length, 24], [Date, Thu, 09 Apr 2026 14:14:56 GMT]}
RawContentLength  : 24



PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-6\part-d-node> docker compose ps
time="2026-04-09T19:45:11+05:30" level=warning msg="C:\\Users\\Tushar Mangal\\Documents\\Containerization and DevOps\\Lab\\Lab-6\\part-d-node\\docker-compos
e.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME             IMAGE                  COMMAND                  SERVICE   CREATED         STATUS         PORTS
custom-node-app   part-d-node-nodeapp   "docker-entrypoint.s…"   nodeapp   53 seconds ago   Up 52 seconds   0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
```

---

15 Experiment 6B: WordPress + MySQL using Compose

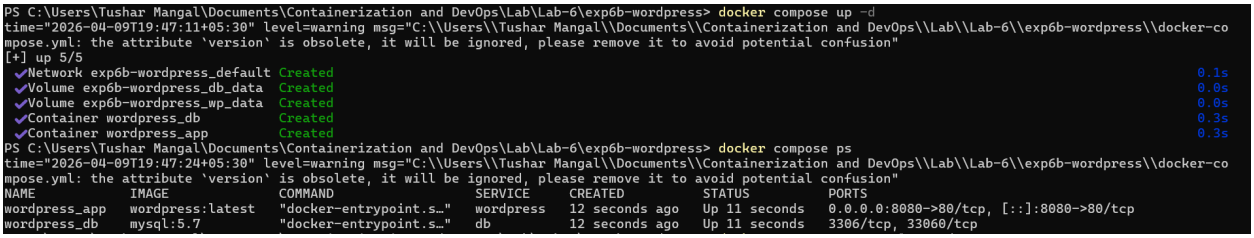
Using a single Compose file completely replaces the manual networking and container execution from Task 2.

docker-compose.yml

```
version: '3.9'
services:
  db:
    image: mysql:5.7
    container_name: wordpress_db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    container_name: wordpress_app
    depends_on:
      - db
    ports:
      - "8080:80"
    restart: always
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wordpress
    volumes:
      - wp_data:/var/www/html

volumes:
  db_data:
  wp_data:
```

15.0.1 1. Run the Stack

```
docker compose up -d
docker ps
```



15.0.2 2. Scaling

```
docker compose up --scale wordpress=3
```

```
PS C:\Users\Tushar Mangal\Documents\Containerization and DevOps\Lab\Lab-6\exp6b-wordpress> docker compose up --scale wordpress=3
time="2026-04-09T19:47:40+05:30" level=warning msg="C:\\Users\\Tushar Mangal\\Documents\\Containerization and DevOps\\Lab\\Lab-6\\exp6b-wordpress\\docker-co
mpose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 1/1
 ✔Container wordpress_db Running                                                                                                              0.0s
WARNING: The "wordpress" service is using the custom container name "wordpress_app". Docker requires each container to have a unique name. Remove the custom
 name to scale the service
```

Important Learning Outcome: The scaling intentionally throws an error because standard Docker Compose is limited here. Since `container_name: wordpress_app` and `ports: "8080:80"` are hardcoded in the file, Compose cannot launch 3 replicas that all want the identical name and host port. This demonstrates the exact scenario where **Docker Swarm** (which handles dynamic naming and load balances identical ports) becomes necessary for production!

15.1 Docker Compose vs Swarm Comparison

Feature	Compose	Swarm
Scope	Single host	Multi-node
Scaling	Manual / Limited by port bindings	Built-in
Load balancing	No	Yes
Self-healing	No	Yes

15.2 Key Takeaways

1. **Compose simplifies multi-container apps** by keeping settings in version control.
2. **Declarative approach** is much easier than manually tracking `docker run` flags.
3. **Compose is excellent for dev/test**, but limited in native scaling due to port/name collisions without strict dynamic tuning.
4. **Docker Swarm** is much better for production cluster orchestration, load balancing, and self-healing.

---

16 Containerization-and-DevOps

17 Lab 7: CI/CD using Jenkins, GitHub & Docker Hub

17.1 1. Aim

To build a complete CI/CD pipeline using:

- **GitHub** — Source code repository
- **Jenkins** — Automation server
- **Docker Hub** — Container image storage

17.2 2. Objective

- Understand the CI/CD workflow end-to-end
- Automate the build and push process using Jenkins
- Write a Jenkins pipeline using `Jenkinsfile`
- Store credentials securely inside Jenkins
- Trigger automated builds using GitHub Webhooks

17.3 3. What is CI/CD?

17.3.1 CI – Continuous Integration

Code is automatically built and tested after every commit pushed to GitHub.

17.3.2 CD – Continuous Deployment

The tested application is automatically packaged and deployed without manual steps.

17.3.3 Flow Diagram

```
Developer → GitHub → Webhook → Jenkins → Docker Hub
```

17.4 4. What is Jenkins?

Jenkins is a **GUI-based open-source automation server** that enables:

- **Building** applications
- **Testing** code automatically
- **Deploying** to target environments

17.4.1 Key Features

Feature	Description
Web Dashboard	Visual interface to manage jobs and pipelines
Plugin Support	1800+ plugins for every major tool
Pipeline as Code	Define your full CI/CD flow in a `Jenkinsfile`
Credential Store	Securely handles secrets and tokens

17.5 5. Project Structure

```
jenkins_lab-main/
    app.py
    requirements.txt
    Dockerfile
    Jenkinsfile
```

17.6 6. Application Code (Flask)

app.py

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def home():
    return "Hello from CI/CD Pipeline!"

app.run(host="0.0.0.0", port=80)
```

requirements.txt

```
flask
```

17.7 7. Dockerfile

```
FROM python:3.10-slim
WORKDIR /app
COPY . .
RUN pip install -r requirements.txt
EXPOSE 80
CMD ["python", "app.py"]
```

17.8 8. Jenkinsfile (Pipeline)

The `Jenkinsfile` defines the full CI/CD pipeline as code:

```
pipeline {
    agent any

    environment {
        IMAGE_NAME = "your-dockerhub-username/myapp"
    }

    stages {
        stage('Clone Source') {
            steps {
                git 'https://github.com/your-username/my-app.git'
            }
        }

        stage('Build Docker Image') {
            steps {
                sh 'docker build -t $IMAGE_NAME:latest .'
            }
        }

        stage('Login to Docker Hub') {
            steps {
                withCredentials([string(credentialsId: 'dockerhub-token', variable: 'DOCKER_TOKEN')]) {
                    sh 'echo $DOCKER_TOKEN | docker login -u your-dockerhub-username --password-stdin'
                }
            }
        }

        stage('Push to Docker Hub') {
            steps {
                sh 'docker push $IMAGE_NAME:latest'
            }
        }
    }
}
```

17.8.1 Pipeline Stages Explained

Stage	What It Does
`Clone Source`	Pulls the latest code from GitHub
`Build Docker Image`	Runs `docker build` to create the image
`Login to Docker Hub`	Authenticates using stored token credential
`Push to Docker Hub`	Uploads the built image to Docker Hub

17.9 9. Jenkins Setup using Docker

We run Jenkins itself as a Docker container for easy setup.

docker-compose.yml

```
version: "3.8"
services:
  jenkins:
    image: jenkins/jenkins:lts
    container_name: jenkins
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock
    user: root

volumes:
  jenkins_home:
```

> **Why mount `/var/run/docker.sock`?** This allows Jenkins (running inside a container) to communicate with the host Docker daemon — so it can build and push images directly from within the pipeline.

17.9.1 Starting Jenkins

```
docker compose up -d
```

Then open: **http://localhost:8080**

17.10 10. Jenkins Configuration

17.10.1 Step 1 – Add Docker Hub Credentials

Navigate to:

`Manage Jenkins → Credentials → System → Global credentials → Add Credentials`

- **Kind:** Secret text
- **ID:** `dockerhub-token`
- **Secret:** Your Docker Hub access token

17.10.2 Step 2 – Create a Pipeline Job

1. Click **New Item**
2. Select **Pipeline**
3. Under **Pipeline Definition**, select `Pipeline script from SCM`
4. Set **SCM** to Git and paste your GitHub repo URL
5. Jenkins will automatically pick up the `Jenkinsfile` from the repo root

---

17.11 11. Webhook Integration

17.11.1 In GitHub

1. Go to your repo → **Settings → Webhooks → Add webhook**
2. Set the Payload URL to:
   `http://<your-jenkins-ip>:8080/github-webhook/`
3. Content type: `application/json`
4. Trigger on: **Push events**

Now every `git push` will automatically trigger the Jenkins pipeline!

17.12 12. Execution Flow

```
Step 1: Developer pushes code to GitHub
         ↓
Step 2: GitHub sends a Webhook notification to Jenkins
         ↓
Step 3: Jenkins Pipeline runs:
        → Clone code
        → Build Docker image
        → Login to Docker Hub
        → Push image
         ↓
Step 4: Image is now live on Docker Hub
```

17.13 13. Key Concept: `withCredentials`

17.13.1 Problem

Hardcoding passwords in `Jenkinsfile` exposes secrets in version control.

17.13.2 Solution

```
withCredentials([string(credentialsId: 'dockerhub-token', variable: 'DOCKER_TOKEN')]) {
    sh 'echo $DOCKER_TOKEN | docker login -u your-dockerhub-username --password-stdin'
}
```

Jenkins:

- Fetches the secret from the encrypted credentials store
- Injects it as an environment variable **temporarily**
- Automatically masks it from all logs
- Removes it from memory after the block completes

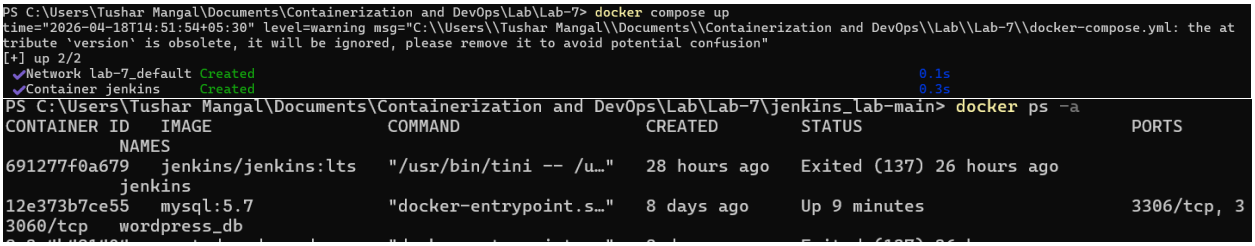
17.14 14. Key Commands in Pipeline

Command	Purpose
`git`	Clone the source code repository
`sh`	Execute a shell command on the agent
`docker build`	Build a Docker image from the `Dockerfile`
`docker login`	Authenticate with Docker Hub
`docker push`	Upload the image to Docker Hub

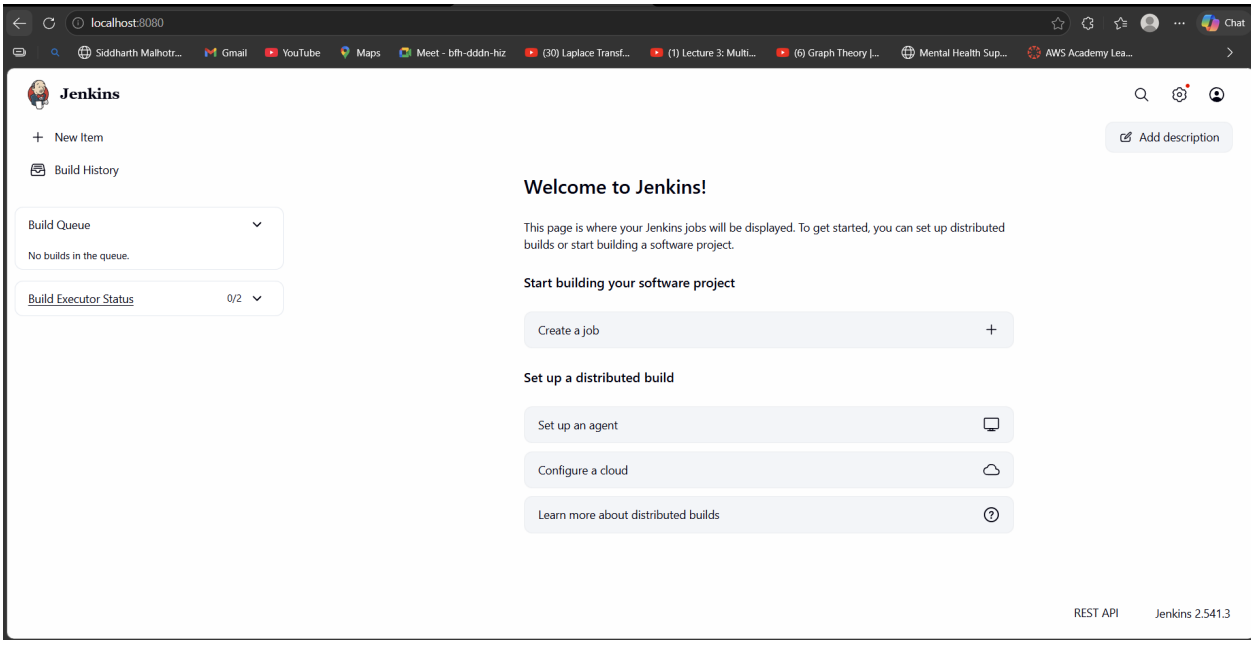
17.15 15. Observations & Screenshots

The following screenshots document the complete setup and execution of this CI/CD pipeline.

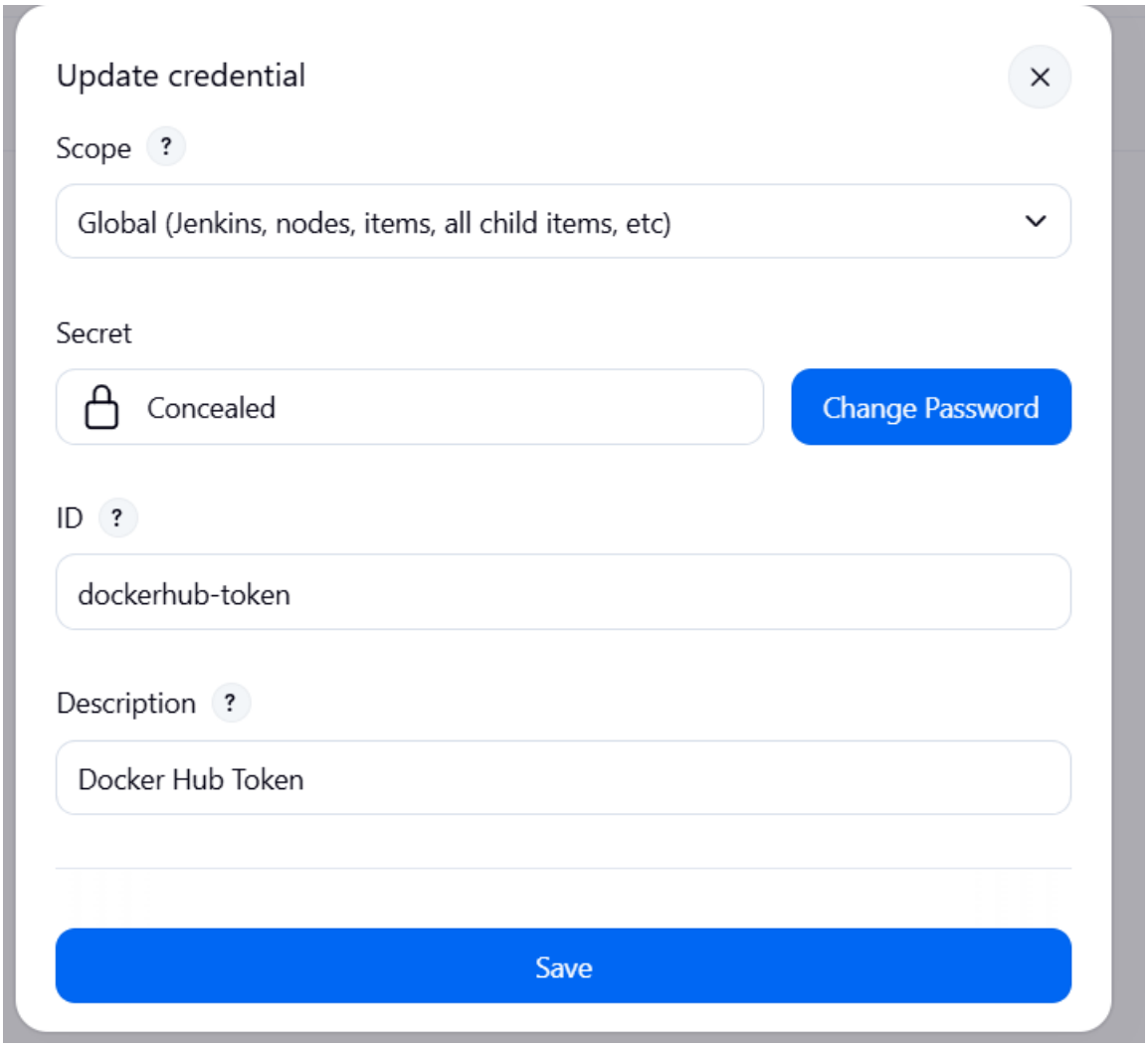
17.15.1 Jenkins Running via Docker Compose



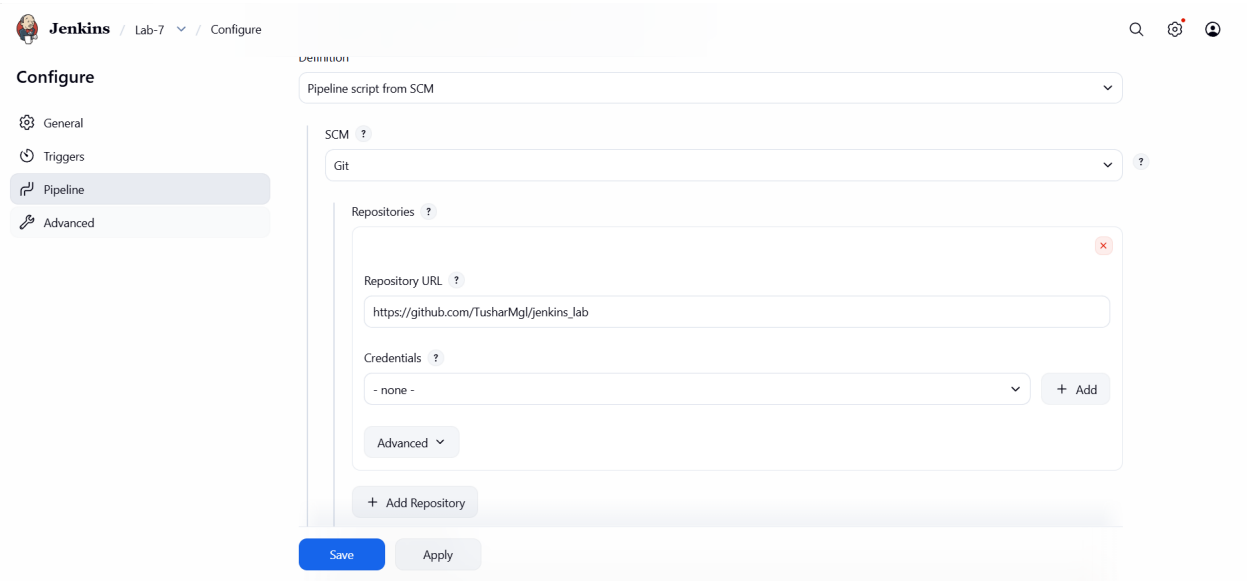
17.15.2 Jenkins Initial Setup & Dashboard



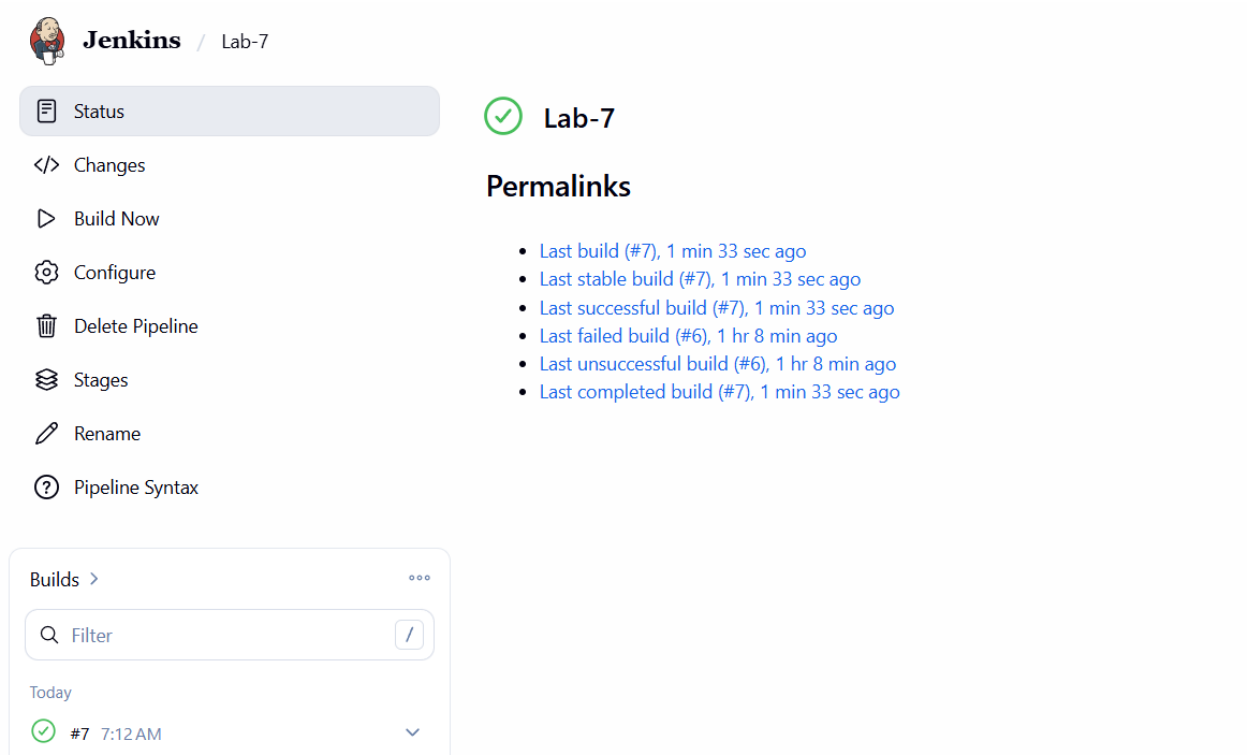
17.15.3 Adding Docker Hub Credentials in Jenkins



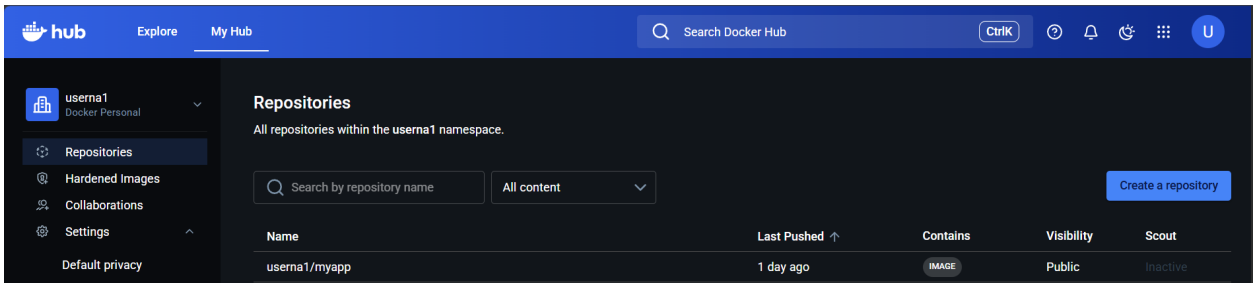
17.15.4 Pipeline Job Configuration



17.15.5 Pipeline Build Execution



17.15.6 Image Successfully Pushed to Docker Hub



17.16

17.17 16. Result

Successfully created a complete automated CI/CD pipeline:

Component	Role
GitHub	Stores source code, triggers webhook on push
Jenkins	Automates build, login, and push stages
Docker	Provides consistent build environment
Docker Hub	Stores and distributes the built image

Every `git push` now **automatically** results in a new Docker image on Docker Hub — with **zero manual steps**.

17.18 Key Takeaways

1. **Jenkins automates** the entire pipeline — no manual intervention needed after setup.
2. **GitHub stores code** and acts as the trigger source via webhooks.
3. **Docker ensures consistency** — the same image runs anywhere.
4. **`withCredentials`** safely handles secrets without exposing them in logs or code.
5. **Pipeline as Code** (`Jenkinsfile`) keeps your CI/CD logic version-controlled alongside your app.

---

18 Containerization-and-DevOps

18.1 Experiment 10: Working with SonarQube

18.1.0.1 Introduction

18.1.0.2 What is SonarQube? SonarQube is and open-source platform for continous inspection of code quality. It perform automatic reviews with static analysis to detect bugs, code smells &/ security vulnerabilities.

18.1.0.3 How SonarQube Solves the Problem:

- **Continous Inspection**: Scans code with everey commit, providing immediate feedback.
- **Quality Gates:** Defines pass/fail criteria for code quality.

- **Technical Debt Quantification:** Measures effort needed to fiz issues.
- **Multi-language Support:** Supports 20+ programming languages.
- **Visual Analytics:** Dashboard showing code quality metrics and trends.

18.1.0.4 Hands-On

Step-1:- Create a `docker-compose.yml`

```
nano docker-compose.yml
```

Paste This:

```
version: '3.8'

services:
  sonar-db:
    image: postgres:13
    container_name: sonar-db
    environment:
      POSTGRES_USER: sonar
      POSTGRES_PASSWORD: sonar
      POSTGRES_DB: sonarqube
    volumes:
      - sonar-db-data:/var/lib/postgresql/data
    networks:
      - sonarqube-lab

  sonarqube:
    image: sonarqube:lts-community
    container_name: sonarqube
    ports:
      - "9000:9000"
    environment:
      SONAR_JDBC_URL: jdbc:postgresql://sonar-db:5432/sonarqube
      SONAR_JDBC_USERNAME: sonar
      SONAR_JDBC_PASSWORD: sonar
    volumes:
      - sonar-data:/opt/sonarqube/data
      - sonar-extensions:/opt/sonarqube/extensions
    depends_on:
      - sonar-db
    networks:
      - sonarqube-lab

volumes:
  sonar-db-data:
  sonar-data:
  sonar-extensions:

networks:
  sonarqube-lab:
    driver: bridge
```

```
tushar_mangal@LAPTOP-1DD
 GNU nano 7.2                                  docker-compose.yml *
version: '3.8'

services:
  sonar-db:
    image: postgres:13
    container_name: sonar-db
    environment:
      POSTGRES_USER: sonar
      POSTGRES_PASSWORD: sonar
      POSTGRES_DB: sonarqube
    volumes:
      - sonar-db-data:/var/lib/postgresql/data
    networks:
      - sonarqube-lab

  sonarqube:
    image: sonarqube:lts-community
    container_name: sonarqube
    ports:
      - "9000:9000"
    environment:
      SONAR_JDBC_URL: jdbc:postgresql://sonar-db:5432/sonarqube
      SONAR_JDBC_USERNAME: sonar
      SONAR_JDBC_PASSWORD: sonar
    volumes:
      - sonar-data:/opt/sonarqube/data
```

Step-2:- Start Compose

`docker-compose up -d`

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-10$ docker-compos
e up -d
WARN[0000] /mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-10/docker-compose.yml: the attribute
 `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 34/34
 ✔Image postgres:13                Pulled                                                                    46.7s
 ✔Image sonarqube:lts-community    Pulled                                                                    66.0s
 ✔Network lab-10_sonarqube-lab     Created                                                                    0.2s
 ✔Volume lab-10_sonar-db-data      Created                                                                    0.0s
 ✔Volume lab-10_sonar-data         Created                                                                    0.0s
 ✔Volume lab-10_sonar-extensions   Created                                                                    0.0s
 ✔Container sonar-db               Created                                                                    1.1s
 ✔Container sonarqube              Created                                                                    0.3s
```

Step-3:- Verify SonarQube is running via Logs

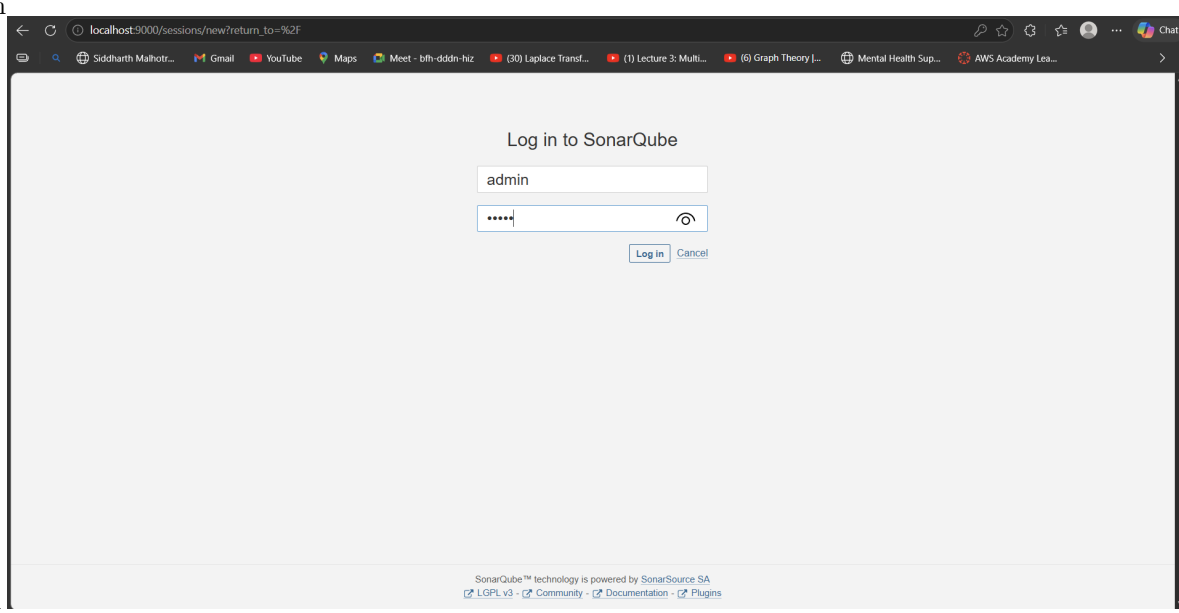
`docker-compose logs -f sonarqube`

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-10$ docker-compos
e logs -f sonarqube
WARN[0000] /mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-10/docker-compose.yml: the attribute
 `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
sonarqube  | 2026.04.25 06:58:56 INFO  app[][o.s.a.AppFileSystem] Cleaning or creating temp directory /opt/sonarqube/tem
p
sonarqube  | 2026.04.25 06:58:56 INFO  app[][o.s.a.es.EsSettings] Elasticsearch listening on [HTTP: 127.0.0.1:9001, TCP:
 127.0.0.1:38845]
sonarqube  | 2026.04.25 06:58:56 INFO  app[][o.s.a.ProcessLauncherImpl] Launch process[ELASTICSEARCH] from [/opt/sonarqu
be/elasticsearch]: /opt/sonarqube/elasticsearch/bin/elasticsearch
sonarqube  | 2026.04.25 06:58:56 INFO  app[][o.s.a.SchedulerImpl] Waiting for Elasticsearch to be up and running
sonarqube  | 2026.04.25 06:58:59 INFO  es[][o.e.n.Node] version[7.17.15], pid[26], build[default/tar/0b8ecfb4378335f4689
c4223d1f1115f16bef3ba/2023-11-10T22:03:46.987399016Z], OS[Linux/6.6.87.2-microsoft-standard-WSL2/amd64], JVM[Eclipse Ado
ptium/OpenJDK 64-Bit Server VM/17.0.16/17.0.16+8]
sonarqube  | 2026.04.25 06:58:59 INFO  es[][o.e.n.Node] JVM home [/opt/java/openjdk]
sonarqube  | 2026.04.25 06:58:59 INFO  es[][o.e.n.Node] JVM arguments [-XX:+UseG1GC, -Djava.io.tmpdir=/opt/sonarqube/tem
p, -XX:ErrorFile=/opt/sonarqube/logs/es_hs_err_pid%p.log, -Des.networkaddress.cache.ttl=60, -Des.networkaddress.cache.ne
gative.ttl=10, -XX:+AlwaysPreTouch, -Xss1m, -Djava.awt.headless=true, -Dfile.encoding=UTF-8, -Djna.nosys=true, -Djna.tmp
dir=/opt/sonarqube/temp, -XX:-OmitStackTraceInFastThrow, -Dio.netty.noUnsafe=true, -Dio.netty.noKeySetOptimization=true,
 -Dio.netty.recycler.maxCapacityPerThread=0, -Dio.netty.allocator.numDirectArenas=0, -Dlog4j.shutdownHookEnabled=false,
-Dlog4j2.disable.jmx=true, -Dlog4j2.formatMsgNoLookups=true, -Djava.locale.providers=COMPAT, -Dcom.redhat.fips=false, -D
es.enforce.bootstrap.checks=true, -Xmx512m, -Xms512m, -XX:MaxDirectMemorySize=256m, -XX:+HeapDumpOnOutOfMemoryError, -De
s.path.home=/opt/sonarqube/elasticsearch, -Des.path.conf=/opt/sonarqube/temp/conf/es, -Des.distribution.flavor=default,
-Des.distribution.type=tar, -Des.bundled_jdk=false]
sonarqube  | 2026.04.25 06:59:00 INFO  es[][o.e.p.PluginsService] loaded module [analysis-common]
sonarqube  | 2026.04.25 06:59:00 INFO  es[][o.e.p.PluginsService] loaded module [lang-painless]
sonarqube  | 2026.04.25 06:59:00 INFO  es[][o.e.p.PluginsService] loaded module [parent-join]
```

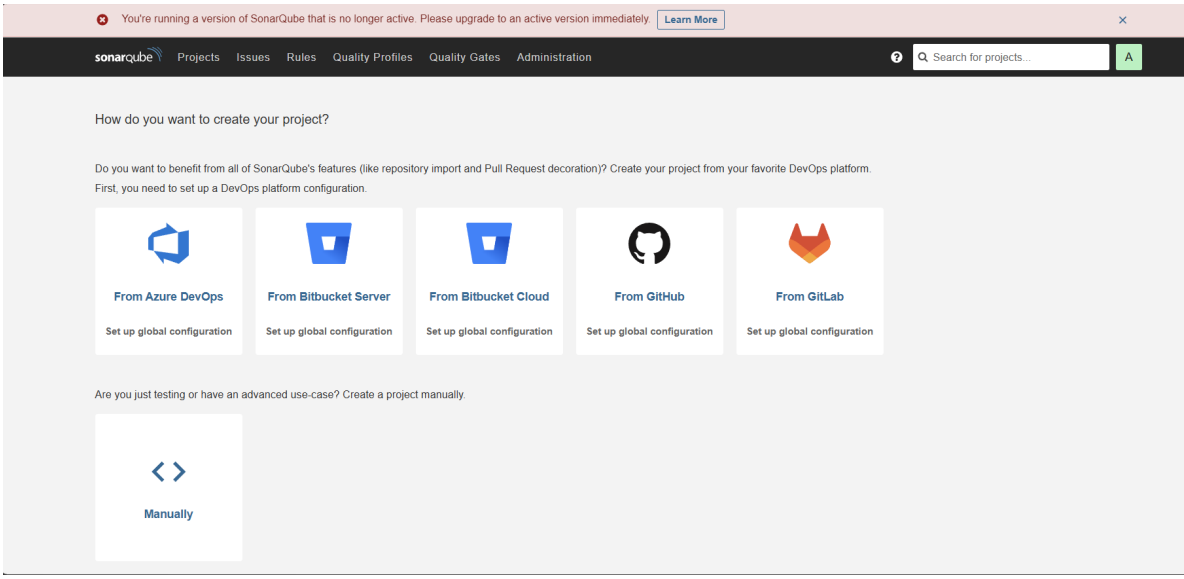
Step-4:- Login SonarQube

`http://localhost:9001`

- username: admin



- password: admin



Step-5:- HomePage

Step-6:- Create Sample Java App

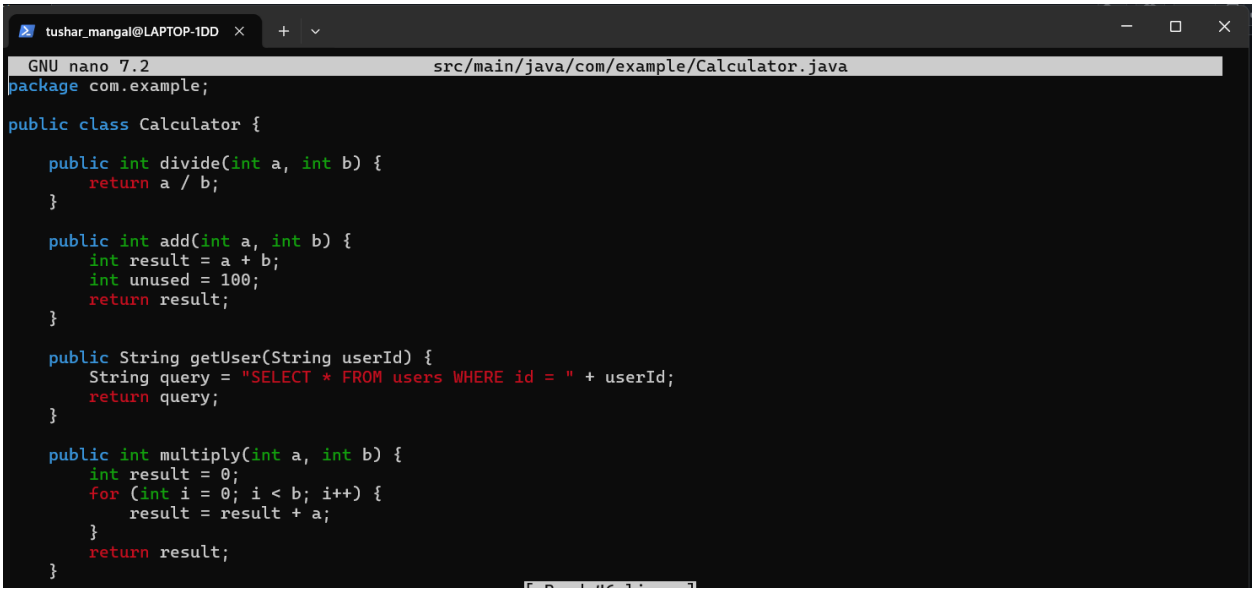
```
mkdir -p sample-java-app/src/main/java/com/example
cd sample-java-app
```

Step-7:- Create `Calculator.java`

```
nano src/main/java/com/example/Calculator.java
```

```
package com.example;

public class Calculator {

    public int divide(int a, int b) {
        return a / b;
    }
```

```
    public int add(int a, int b) {
        int result = a + b;
        int unused = 100;
        return result;
    }

    public String getUser(String userId) {
        String query = "SELECT * FROM users WHERE id = " + userId;
        return query;
    }

    public int multiply(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }

    public int multiplyAlt(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }

    public String getName(String name) {
        return name.toUpperCase();
    }

    public void riskyOperation() {
        try {
            int x = 10 / 0;
        } catch (Exception e) {
        }
    }
}
```



Step-8:- Create pom.xml

```
nano pom.xml
```

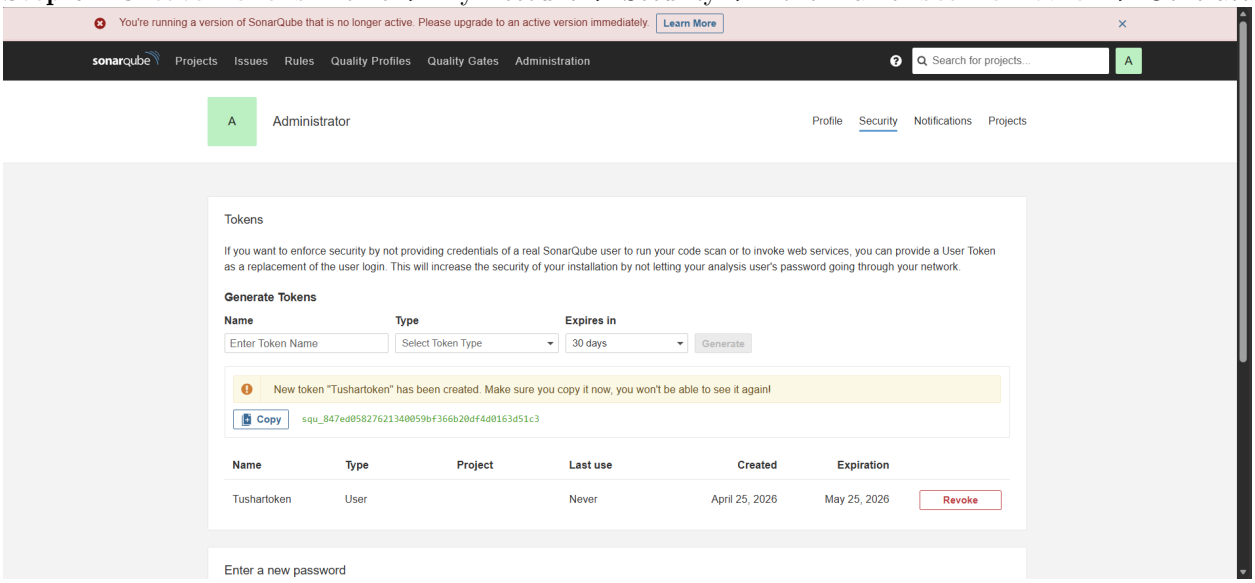
```
<project xmlns="http://maven.apache.org/POM/4.0.0">
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>sample-app</artifactId>
    <version>1.0-SNAPSHOT</version>

    <properties>
        <sonar.projectKey>sample-java-app</sonar.projectKey>
        <sonar.host.url>http://localhost:9000</sonar.host.url>
        <sonar.login>YOUR_TOKEN</sonar.login>
    </properties>

    <build>
        <plugins>
            <plugin>
                <groupId>org.sonarsource.scanner.maven</groupId>
                <artifactId>sonar-maven-plugin</artifactId>
                <version>3.9.1.2184</version>
            </plugin>
        </plugins>
    </build>
</project>
```

tushar_mangal@LAPTOP-1DD

GNU nano 7.2 pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0">
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>sample-app</artifactId>
    <version>1.0-SNAPSHOT</version>

    <properties>
        <sonar.projectKey>sample-java-app</sonar.projectKey>
        <sonar.host.url>http://localhost:9000</sonar.host.url>
        <sonar.login>YOUR_TOKEN</sonar.login>
    </properties>

    <build>
        <plugins>
            <plugin>
                <groupId>org.sonarsource.scanner.maven</groupId>
                <artifactId>sonar-maven-plugin</artifactId>
                <version>3.9.1.2184</version>
            </plugin>
        </plugins>
    </build>
</project>
```

Step-9:- Create Tokens Profile -> My Account -> Security -> TokenName: `scanner-token` -> Generate



Step-10:- Run Scanner

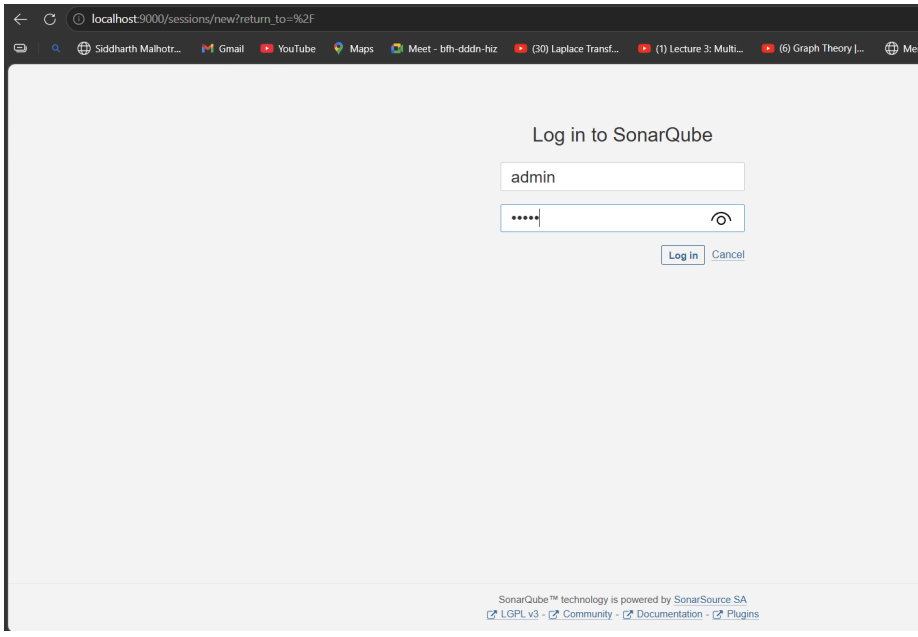
```
mvn sonar:sonar -Dsonar.login=YOUR_TOKEN
```

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-10/sample-java-ap
p$ mvn sonar:sonar -Dsonar.login=squ_847ed0582762134005 9bf366b20df4d0163d51c3
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< com.example:sample-app >----------------------
[INFO] Building sample-app 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- sonar-maven-plugin:3.9.1.2184:sonar (default-cli) @ sample-app ---
[INFO] User cache: /home/tushar_mangal/.sonar/cache
[INFO] SonarQube version: 9.9.8.100196
[INFO] Default locale: "en", source code encoding: "UTF-8" (analysis is platform dependent)
[INFO] Load global settings
[INFO] Load global settings (done) | time=192ms
[INFO] Server id: 74C15348-AZ3Db49ZBNtJ7P7afIFZ
[INFO] User cache: /home/tushar_mangal/.sonar/cache
[INFO] Load/download plugins
[INFO] Load plugins index
[INFO] Load plugins index (done) | time=118ms
[INFO] Load/download plugins (done) | time=4240ms
[INFO] Process project properties
[INFO] Process project properties (done) | time=28ms
[INFO] Execute project builders
```

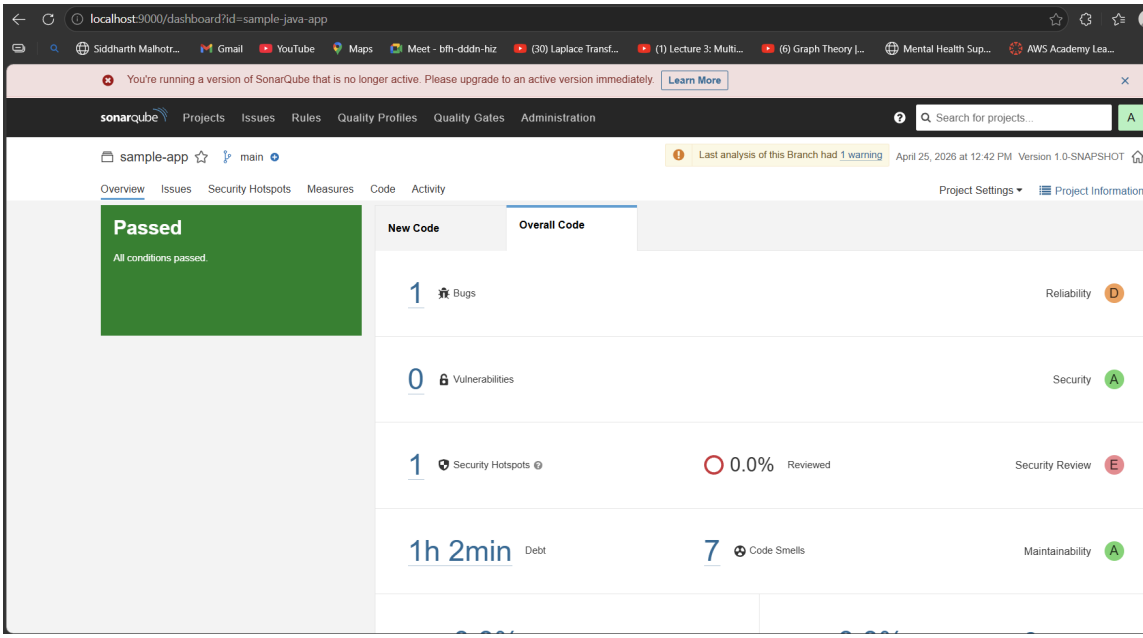
Step-11:- Scroll down to view Build Success

```
[WARNING] Thread[#50,JGit-FileStoreAttributeReader-1,5,main]: got smaller file timest
ushar Mangal/Documents/Containerization and DevOps/Lab/.git: 2026-04-25T07:12:27Z < 2
g measurement at resolution PT0.0154685S.
[INFO] SCM Publisher 0/2 source files have been analyzed (done) | time=673ms
[WARNING] Missing blame information for the following files:
[WARNING]   * src/main/java/com/example/Calculator.java
[WARNING]   * pom.xml
[WARNING] This may lead to missing/broken features in SonarQube
[INFO] CPD Executor Calculating CPD for 1 file
[INFO] CPD Executor CPD calculation finished (done) | time=14ms
[INFO] Analysis report generated in 238ms, dir size=127.8 kB
[INFO] Analysis report compressed in 1056ms, zip size=19.2 kB
[INFO] Analysis report uploaded in 429ms
[INFO] ANALYSIS SUCCESSFUL, you can find the results at: http://localhost:9000/dashbo
[INFO] Note that you will be able to access the updated dashboard once the server has
eport
[INFO] More about the report processing at http://localhost:9000/api/ce/task?id=AZ3De
[INFO] Analysis total time: 13.836 s
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  20.331 s
[INFO] Finished at: 2026-04-25T07:12:30Z
[INFO] ------------------------------------------------------------------------
```

Step-12:- On Browser test will be Listed



Step-13:- View Report



---

19 Containerization-and-DevOps

19.1 Experiment 11: Orchestration via Docker Swarm

19.1.0.1 Introduction

19.1.0.2 The Progression Path

`docker run` → `Docker Compose` → `Docker Swarm` → `Kubernetes`

Single container	Multi-container (single host)	Orchestration (basic)	Advanced orchestration

This experiment focuses on: Moving from Compose → Swarm

19.1.0.3 Hands-On

Step-1:- Create a `docker-compose.yml`

```
nano docker-compose.yml
```

Paste This:

```
version: '3.9'

services:
  db:
```

```
    image: mysql:5.7
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    depends_on:
      - db
    ports:
      - "8080:80"
    restart: always
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wordpress
    volumes:
      - wp_data:/var/www/html

volumes:
  db_data:
  wp_data:
```

Step-2:- Initialize Swarm

```
docker swarm init
```

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-11$ docker swarm
init
Swarm initialized: current node (lg5xjls2gvcegkvi7lxdeywj5) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-2vuhfoic12j1wnav4gelksw0zu7g64kxn57vo335rbixl20k3y-1rr5m0ol0o3ubavvj799qg3vn 192.
168.65.3:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
```

Step-3:- Verify Swarm is Active

`docker node ls`

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-11$ docker node l
s
ID                            HOSTNAME         STATUS    AVAILABILITY   MANAGER STATUS   ENGINE VERSION
lg5xjls2gvcegkvi7lxdeywj5 *   docker-desktop   Ready     Active         Leader           29.2.1
```

Step-4:- Deploy as a stack

`docker stack deploy -c docker-compose.yml wpstack`

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-11$ docker stack
deploy -c docker-compose.yml wpstack
Ignoring unsupported options: restart

Since --detach=false was not specified, tasks will be created in the background.
In a future release, --detach=false will become the default.
Creating network wpstack_default
Creating service wpstack_db
Creating service wpstack_wordpress
```

Step-5:- Verify by listing Services

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/La
e ls
ID             NAME              MODE         REPLICAS   IMAGE              PORTS
wcfbhu3skuc9   wpstack_db        replicated   1/1        mysql:5.7
yo5b4ru2uvtf   wpstack_wordpress   replicated   1/1        wordpress:latest   *:8080->80/tcp
```

Step-6:- Details for container service

`docker service ps wpstack_wordpress`

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-11$ docker servic
e ps wpstack_wordpress
ID             NAME                  IMAGE              NODE             DESIRED STATE   CURRENT STATE            ERROR
    PORTS
f64n7jttlx0c   wpstack_wordpress.1   wordpress:latest   docker-desktop   Running         Running 13 seconds ago
```

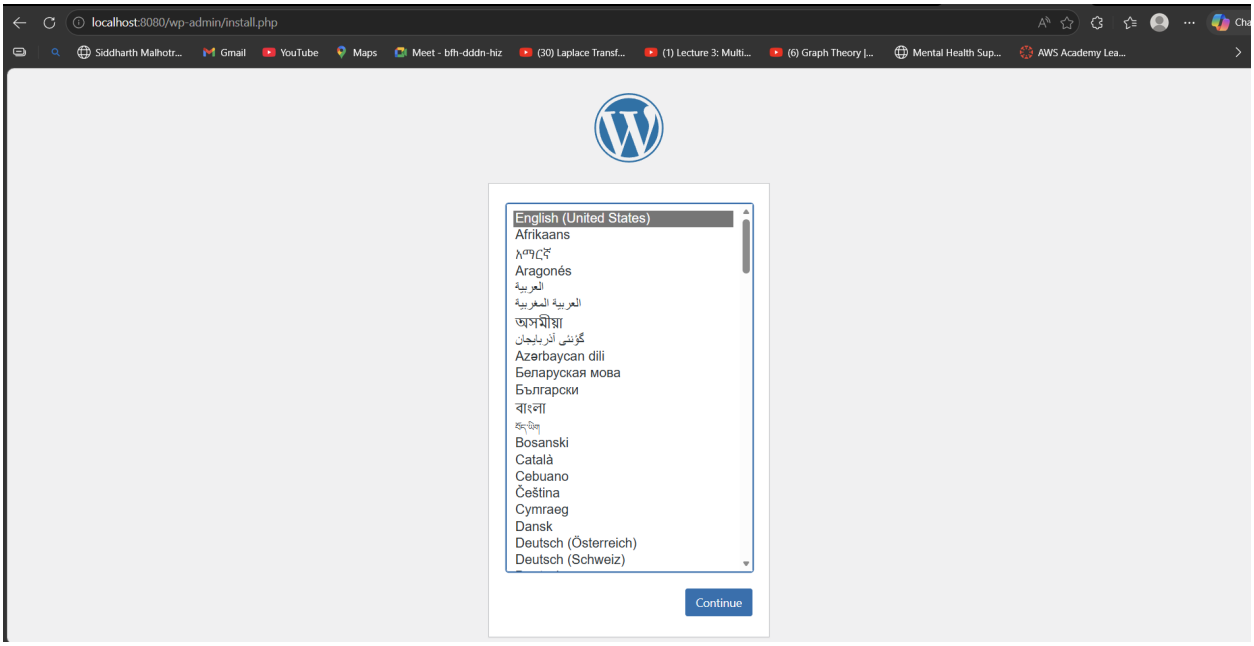
Step-7:- List Containers

`docker ps`

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-11$ docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS                                  PORTS
                                                                                NAMES
7c4250c498f8   wordpress:latest      "docker-entrypoint.s…"   21 seconds ago   Up 20 seconds                           80/tcp
                                                                                wpstack_wordpress.1.f64n7jttlx0c7g
bi3c0km9uux
5859ea75c5a8   mysql:5.7             "docker-entrypoint.s…"   26 seconds ago   Up 25 seconds                           3306/tcp,
 33060/tcp                                                                      wpstack_db.1.w32pfajxaeuxjjqjw9uv3
yyur
f8e585aba56c   jenkins/jenkins:lts   "/usr/bin/tini -- /u…"   12 days ago      Up 11 minutes                           0.0.0.0:5
0000->50000/tcp, [::]:50000->50000/tcp, 0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp   jenkins
12e373b7ce55   mysql:5.7             "docker-entrypoint.s…"   2 weeks ago      Up 11 minutes                           3306/tcp,
 33060/tcp                                                                      wordpress_db
7bbd239ea7c8   node:18-alpine        "docker-entrypoint.s…"   2 weeks ago      Restarting (0) 21 seconds ago
                                                                                webapp
```

Step-8:- Verify application on Containers

`http://localhost:8080/`



Step-9:- Scale application Containers

```
docker service scale wpstack_wordpress=3
```

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-11$ docker servic
e scale wpstack_wordpress=3
wpstack_wordpress scaled to 3
overall progress: 3 out of 3 tasks
1/3: running   [==================================================>]
2/3: running   [==================================================>]
3/3: running   [==================================================>]
verify: Service wpstack_wordpress converged
```

Step-10:- Verify Scalling

```
docker service ls
```

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-11$ docker servic
e ls
ID             NAME                MODE         REPLICAS   IMAGE              PORTS
wcfbhu3skuc9   wpstack_db          replicated   1/1        mysql:5.7
yo5b4ru2uvtf   wpstack_wordpress   replicated   3/3        wordpress:latest   *:8080->80/tcp
```

Step-11:- List Wordpress Containers

```
docker ps | grep wordpress
```

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-11$ docker ps | g
rep wordpress
50444b113059   wordpress:latest      "docker-entrypoint.s…"   18 seconds ago       Up 17 seconds                  80/tc
p                                                                                    wpstack_wordpress.3.zka8k8khz9
q67h0b33mjum876
64658cb019fe   wordpress:latest      "docker-entrypoint.s…"   18 seconds ago       Up 17 seconds                  80/tc
p                                                                                    wpstack_wordpress.2.ibzh08ccge
nxlt5bkzpr2zsq2
7c4250c498f8   wordpress:latest      "docker-entrypoint.s…"   About a minute ago   Up About a minute              80/tc
p                                                                                    wpstack_wordpress.1.f64n7jttlx
0c7gbi3c0km9uux
12e373b7ce55   mysql:5.7             "docker-entrypoint.s…"   2 weeks ago          Up 12 minutes                  3306/
tcp, 33060/tcp                                                                       wordpress_db
```

Step-12:- Kill any Container

```
docker kill <container-id>
```

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-11$ docker kill 5
0444b113059
50444b113059
```

Step-13:- Watch Swarm Recreating Contrainer

```
docker service ps wpstack_wordpress
```

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-11$ docker servic
e ps wpstack_wordpress
ID             NAME                    IMAGE              NODE             DESIRED STATE   CURRENT STATE
      ERROR                         PORTS
f64n7jttlx0c   wpstack_wordpress.1      wordpress:latest   docker-desktop   Running         Running 2 minutes ago

ibzh08ccgenx   wpstack_wordpress.2      wordpress:latest   docker-desktop   Running         Running about a minute ago

ar0l9c6u5age   wpstack_wordpress.3      wordpress:latest   docker-desktop   Running         Running less than a second
ago
```

Step-14:- Verify Container running

```
docker ps | grep wordpress
```

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-11$ docker ps | g
rep wordpress
62dc0b685c5e   wordpress:latest      "docker-entrypoint.s…"   18 seconds ago       Up 12 seconds                  80/tc
p                                                                                   wpstack_wordpress.3.ar0l9c6u5a
ge3xdysilmykvy1
64658cb019fe   wordpress:latest      "docker-entrypoint.s…"   About a minute ago   Up About a minute              80/tc
p                                                                                   wpstack_wordpress.2.ibzh08ccge
nxlt5bkzpr2zsq2
7c4250c498f8   wordpress:latest      "docker-entrypoint.s…"   2 minutes ago        Up 2 minutes                   80/tc
p                                                                                   wpstack_wordpress.1.f64n7jttlx
0c7gbi3c0km9uux
12e373b7ce55   mysql:5.7             "docker-entrypoint.s…"   2 weeks ago          Up 13 minutes                  3306/
tcp, 33060/tcp                                                                      wordpress_db
```

Step-15:- Remove Stack

```
docker stack rm wpstack
```

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-11$ docker stack
rm wpstack
Removing service wpstack_db
Removing service wpstack_wordpress
Removing network wpstack_default
```

Step-16:- Verify Service Removal

```
docker service ls
docker ps
```

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-11$ docker servic
e ls
ID        NAME      MODE      REPLICAS   IMAGE     PORTS
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-11$ docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED       STATUS                                  PORTS
                                                                                 NAMES
f8e585aba56c   jenkins/jenkins:lts   "/usr/bin/tini -- /u…"   12 days ago   Up 14 minutes                           0.0.0.0:5000
0->50000/tcp, [::]:50000->50000/tcp, 0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp   jenkins
12e373b7ce55   mysql:5.7             "docker-entrypoint.s…"   2 weeks ago   Up 14 minutes                           3306/tcp, 33
060/tcp                                                                          wordpress_db
7bbd239ea7c8   node:18-alpine        "docker-entrypoint.s…"   2 weeks ago   Restarting (0) 46 seconds ago
                                                                                 webapp
```

19.1.0.4 Conclusion

19.1.0.5 Summary

You started with	You can now do
Single container (`docker run`)	Multi-container (Compose)
Manual scaling	One-command scaling (`scale`)
Manual recovery	Automatic self-healing
Single host	Multi-host cluster ready

19.1.0.6 Final Takeaway

Compose defines the application. Swarm runs it reliably.

19.1.0.7 Quick Reference Card

```
# Initialize Swarm
docker swarm init

# Deploy stack
docker stack deploy -c docker-compose.yml <stack-name>

# List services
docker service ls

# Scale service
docker service scale <stack-name_service-name>=<replicas>

# See service tasks
docker service ps <service-name>

# Remove stack
docker stack rm <stack-name>

# Leave Swarm (if needed)
docker swarm leave --force
```

---

20 Containerization-and-DevOps

20.1 Experiment 12: Kubernetes

20.1.0.1 Introduction

20.1.0.2 Why Kubernetes over Docker Swarm?

Reason	Explanation
Industry standard	Most companies use Kubernetes
Powerful scheduling	Automatically decides where to run your app
Large ecosystem	Many tools and plugins available
Cloud-native support	Works on AWS, Google Cloud, Azure, etc.

20.1.0.3 Core Kubernetes Concepts (Simple Explanation)

Docker Concept	Kubernetes Equivalent	What it means
Container	**Pod**	A pod is a group of one or more containers. Smallest unit in K8s.

Docker Concept	Kubernetes Equivalent	What it means
Compose service	**Deployment**	Describes how your app should run (e.g., 2 copies, which image to use)
Load balancing	**Service**	Exposes your app to the outside world or other pods
Scaling	**ReplicaSet**	Ensures a certain number of pod copies are always running

---

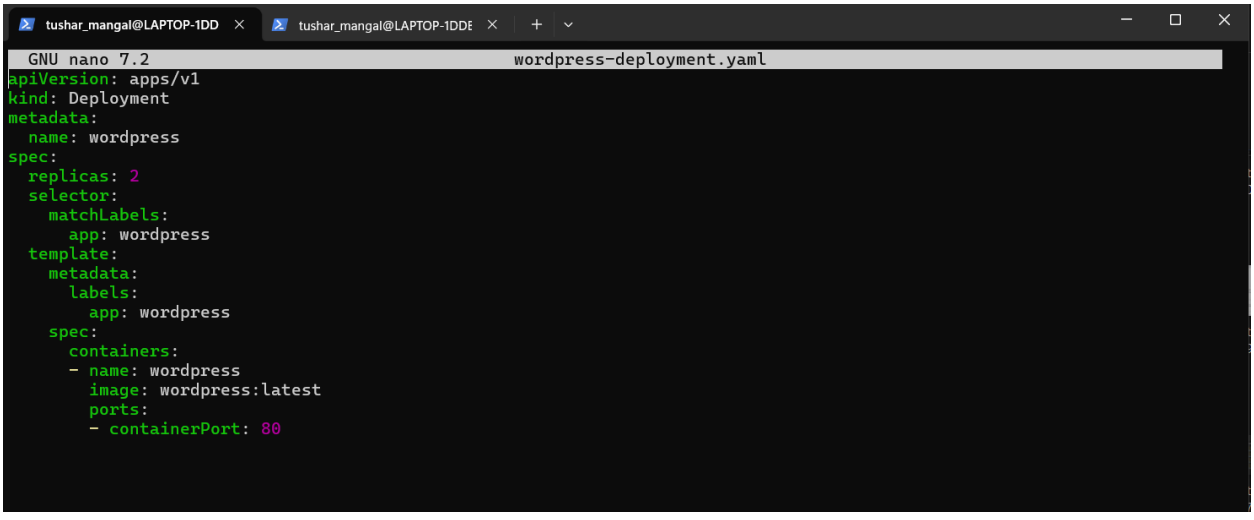
20.1.0.4 Hands-On

Step-1:- Create a `wordpress-deployment.yaml`

```
nano wordpress-deployment.yaml
```

Paste This:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: wordpress
spec:
  replicas: 2
  selector:
    matchLabels:
      app: wordpress
  template:
    metadata:
      labels:
        app: wordpress
    spec:
      containers:
      - name: wordpress
        image: wordpress:latest
        ports:
        - containerPort: 80
```



Step-2:- Apply Deployment

```
kubectl apply -f wordpress-deployment.yaml
```

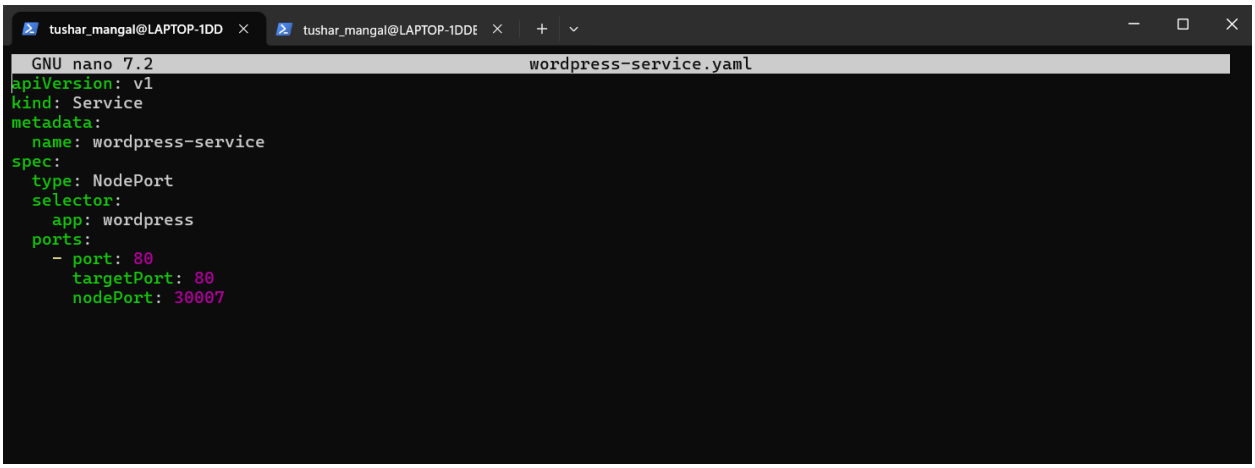
```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-12$ kubectl apply
 -f wordpress-deployment.yaml
deployment.apps/wordpress created
```

Step-3:- Create wordpress-service.yaml

```
nano wordpress-service.yaml
```

Paste this:-

```
apiVersion: v1
kind: Service
metadata:
  name: wordpress-service
spec:
  type: NodePort
  selector:
    app: wordpress
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30007
```



Step-4:- Deploy Service

```
kubectl apply -f wordpress-service.yaml
```

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-12$ kubectl apply
 -f wordpress-service.yaml
service/wordpress-service created
```

Step-5:- Get pods

```
kubectl get pods
```

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-12$ kubectl get p
ods
NAME                         READY   STATUS    RESTARTS   AGE
wordpress-848c87c44f-djdnp   1/1     Running   0          83s
wordpress-848c87c44f-srbmb   1/1     Running   0          83s
```

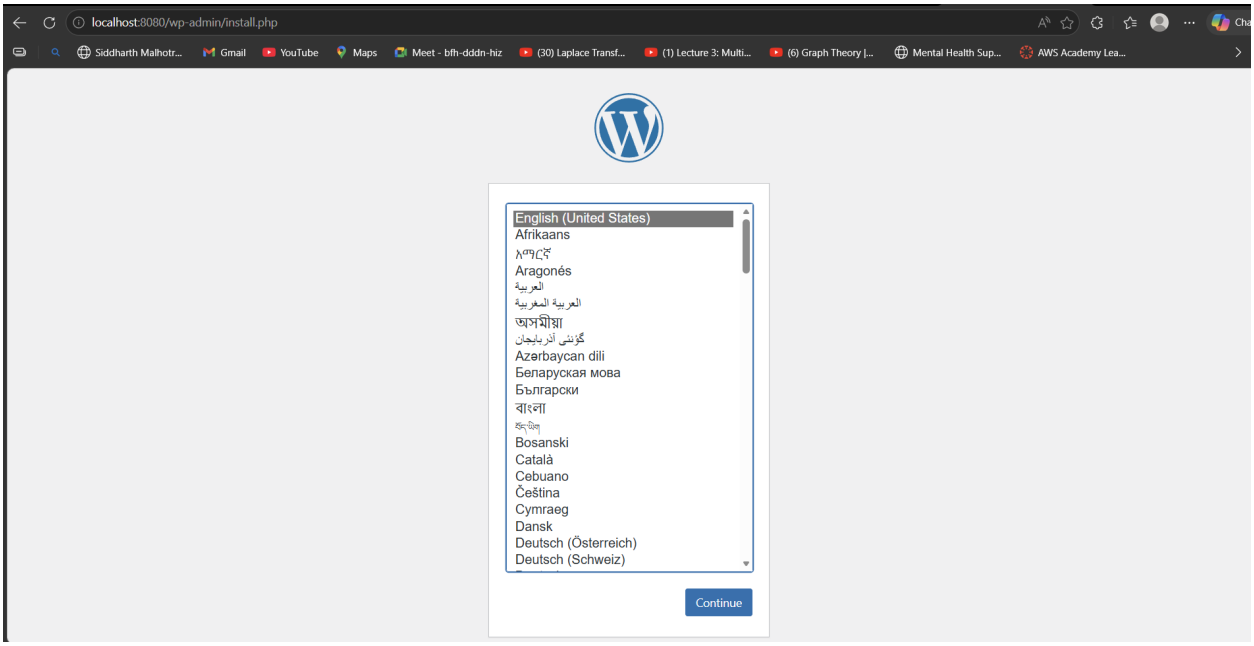
Step-6:- Check service

```
kubectl get svc
```

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-12$ kubectl get s
vc
NAME                TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kubernetes          ClusterIP   10.96.0.1       <none>        443/TCP        2m26s
wordpress-service   NodePort    10.111.48.226   <none>        80:30007/TCP   19s
```

Step-7:- Verify On Browser

`http://localhost:8080/`



Step-8:- Scale Deployment

`kubectl scale deployment wordpress --replicas=4`

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-12$ kubectl scale
 deployment wordpress --replicas=4
deployment.apps/wordpress scaled
```

Step-9:- Get Pods

`kubectl get pods`

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-12$ kubectl get p
ods
NAME                         READY   STATUS              RESTARTS   AGE
wordpress-848c87c44f-7mzmk   1/1     Running             0          6s
wordpress-848c87c44f-djdnp   1/1     Running             0          2m15s
wordpress-848c87c44f-knb27   0/1     ContainerCreating   0          6s
wordpress-848c87c44f-srbmb   1/1     Running             0          2m15s
```

Step-10:- Delete Pods to check Self-healing

`kubectl delete pod <pod-name>`
`kubectl get pods`

```
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-12$ kubectl delet
e pod wordpress-848c87c44f-7mzmk
pod "wordpress-848c87c44f-7mzmk" deleted from default namespace
tushar_mangal@LAPTOP-1DDBCFLP:/mnt/c/Users/Tushar Mangal/Documents/Containerization and DevOps/Lab/Lab-12$ kubectl get p
ods
NAME                         READY   STATUS    RESTARTS   AGE
wordpress-848c87c44f-58fkg   1/1     Running   0          16s
wordpress-848c87c44f-djdnp   1/1     Running   0          3m4s
wordpress-848c87c44f-knb27   1/1     Running   0          55s
wordpress-848c87c44f-srbmb   1/1     Running   0          3m4s
```

20.1.0.5 Insights

20.1.0.6 When to Use Which Tool

Tool	Best for
k3d	Quick learning on your laptop
Minikube	Single-node cluster testing
kubeadm	Real, production-style cluster

20.1.0.7 Summary of Commands (Cheat Sheet)

Goal	Command
Apply a YAML file	`kubectl apply -f file.yaml`
See all pods	`kubectl get pods`
See all services	`kubectl get svc`
Scale a deployment	`kubectl scale deployment <name> --replicas=N`
Delete a pod	`kubectl delete pod <pod-name>`
See all nodes	`kubectl get nodes`
